V-Imp main (Adv)

14/4/21

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Gravitation.

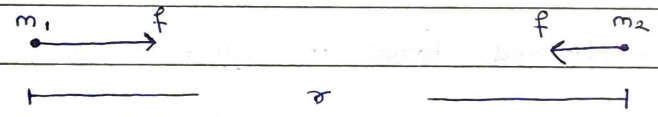
① attractive force by virtue of mass.

② attractive.

f - gravitational force.

↳ An attractive force by virtue of masses is known as gravitational force.

■ 1. Gravitational forces between point masses :-



$f \propto m_1 m_2$

$f \propto \frac{1}{r^2}$ (inverse square law).

$f = \frac{G m_1 m_2}{r^2}$ (experimental law, inverse square law)

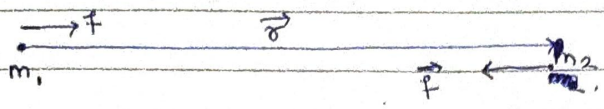
(G): universal gravitational const.

$= 6.67 \times 10^{-11} \frac{N \cdot m^2}{kg^2} = \frac{20}{3} \times 10^{-11} \frac{N \cdot m^2}{kg^2}$

↳ independend of medium.

↳ dimention - $\frac{[MLT^{-2}][L^2]}{[M^2]} = [M^{-1} L^3 T^{-2}]$

1.1 In vector form :-



$$\vec{f} = \frac{G m_1 m_2}{r^2} (-\hat{r})$$

$$= -\frac{G m_1 m_2}{r^3} \vec{r}$$

tail : at the mass due to which $\vec{f}$ is to be found.

Head : at the mass on which $\vec{f}$ is to be found.

$q_1, q_2 \longrightarrow m_1, m_2$.

$K \longrightarrow (-G)$.

Q-1

1 kg

$(0, 2, 3)$ m

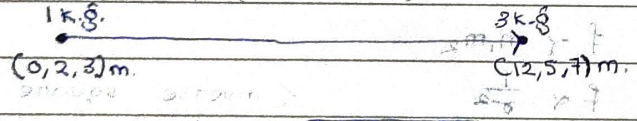
3 kg

$(12, 5, 7)$ m.

find gravitational force in vector form on 3 kg.

Soln-

$\vec{r} = (12\hat{i} + 3\hat{j} + 4\hat{k})$ m.



$$|\vec{r}| = r = \sqrt{12^2 + 3^2 + 4^2}$$

$$= 13 \text{ m}.$$

$$\vec{f} = -\frac{G m_1 m_2}{r^3} \vec{r} = -\frac{20 \times 10^{-11}}{3} \frac{\text{N-m}^2}{\text{kg}^2} \times 1\text{kg} \times 3\text{kg} \times (12\hat{i} + 3\hat{j} + 4\hat{k}) \text{ m}.$$

$$(13\text{m})^3$$

$$= \quad \text{N}.$$

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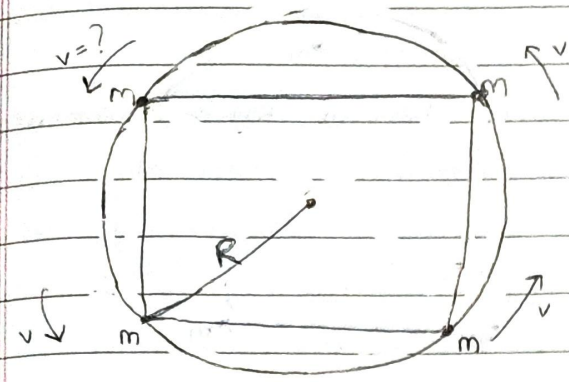
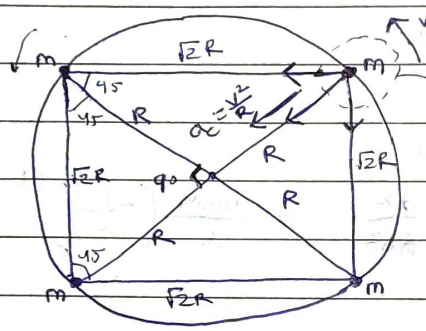
7.2

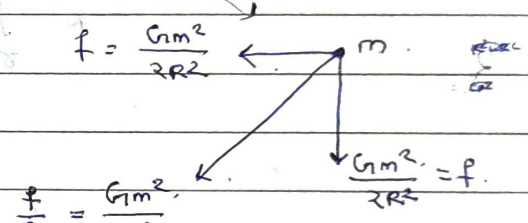


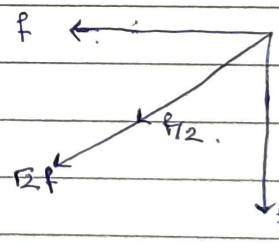
4 identical stars are performing ucm due to gravitational force between them.

Find: ① v (orbital speed).
② Angular velocity
③ time period.

Soln By symmetry :-







$$\Rightarrow f_{net} = \sqrt{2}f + f/2.$$

$$= \left(\sqrt{2} + \tfrac{1}{2}\right)\frac{Gm^2}{2R^2}$$

(i) by NLM, $\left(\sqrt{2} + \frac{1}{2}\right)\frac{Gm^2}{2R^2} = \frac{mv^2}{R}$.

$$v = \sqrt{\left(\sqrt{2} + \frac{1}{2}\right)\frac{Gm}{2R}}$$

(ii) $\omega = \frac{v}{R} = \sqrt{\left(\sqrt{2}+\tfrac{1}{2}\right)\frac{Gm}{2R^3}}$.

(iii) $T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\left(\sqrt{2}+\frac{1}{2}\right)\frac{Gm}{2R^3}}}$

mass per unit length.

Q3

$\lambda = \frac{M}{L}$ M, L (uniform rod)

find Gravitational force between Rod and point masses.

Soln:-

$$dF = \frac{G m \frac{M}{L} dx}{x^2}$$

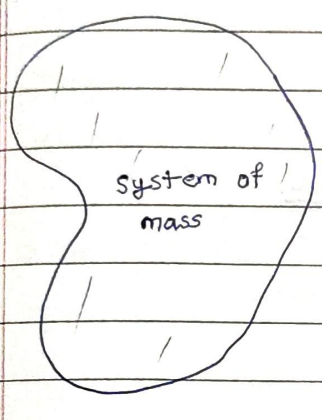
$$F = \frac{GmM}{L} \int_{r}^{r+L} \frac{dx}{dx^2} = \frac{GmM}{L} \left(\frac{-1}{x}\right)_{r}^{r+L}$$

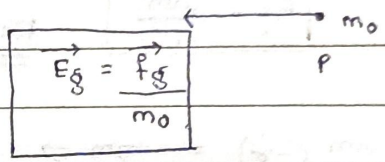
$$= \frac{GmM}{L} \left\langle \frac{1}{r} - \frac{1}{r+L} \right\rangle$$

$$= \frac{GmM}{L} \cdot \frac{L}{r(r+L)} = \frac{GMm}{r(r+L)}$$

Gravitational field intensity :-

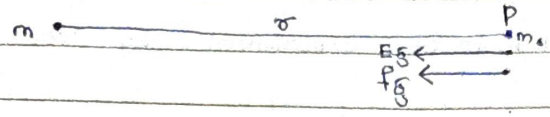
1. Gravitational field intensity at 'P' due to system of masses is defined as gravitation force experienced by test mass per unit test mass.





2. vecton quantity.
3. $\vec{f_e} = m_o \cdot \vec{E_g}$
4. $\vec{f_e} \parallel \vec{E_g}$
5. unit : $\frac{N}{kg} = m/sec^2$
6. dim : $\frac{M L T^{-2}}{M} = [LT^{-2}]$

Gravitational field intensity due to point mass :-

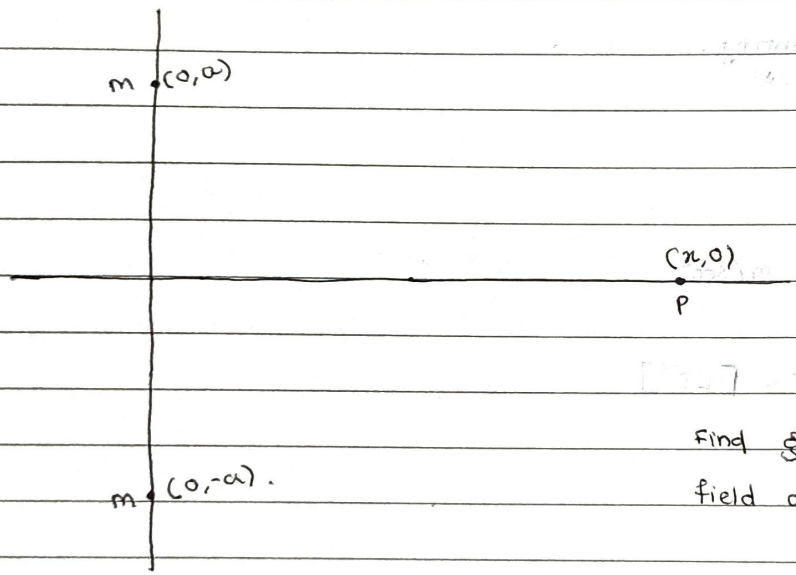


$$E_g = \frac{f_g}{m_o} = \frac{Gmm_o}{r^2 m_o}$$

$$\boxed{E_g = \frac{Gm}{r^2}}$$ Radially in ward.

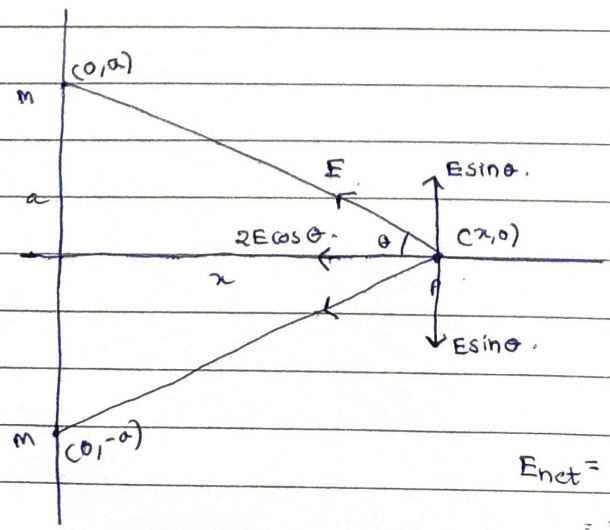
$$\vec{E}_g = \frac{Gm}{r^2}(-\hat{r}) = -\frac{Gm}{r^3}\vec{r}$$

(Q-4)



Find gravitational field at point P.

Soln-



$$E = \frac{GM}{x^2+a^2}.$$

$$E_{net} = 2E\cos\theta.$$

$$= 2 \times \frac{Gm}{a^2+x^2} \frac{x}{\sqrt{a^2+x^2}}$$

$$E_g = \frac{2GMx}{(a^2+x^2)^{3/2}}$$

$x = 0 \Rightarrow E_g = 0$

$x \to \infty \Rightarrow E_g \to 0.$

Emax

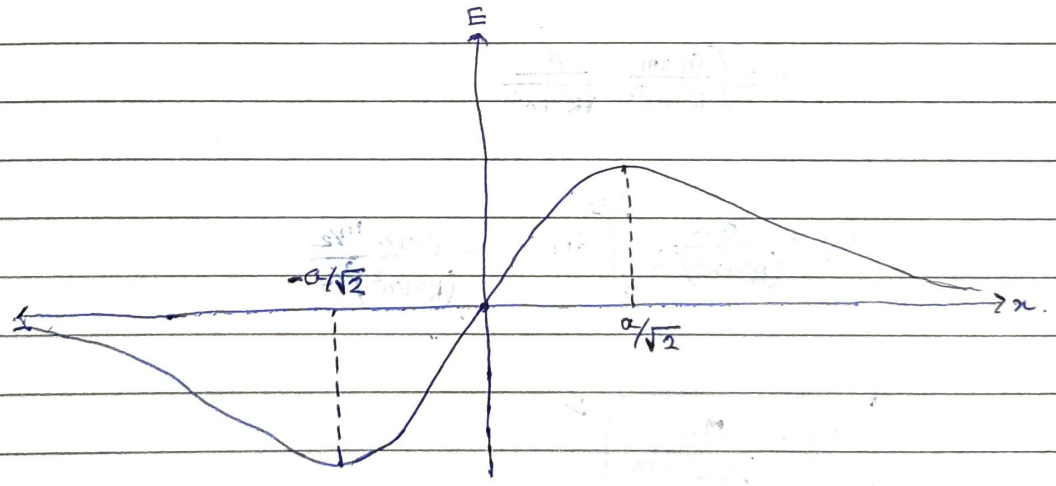
$$\frac{dE}{dx} = 0$$

$$\frac{2GM\,(a^2+x^2)^{3/2}\cdot 1 - \frac{3}{2}(a^2+x^2)^{1/2}\,2x^2}{(a^2+x^2)^3}$$

$$(a^2+x^2)^{3/2} = \frac{3}{2}(a^2+x^2)^{1/2}\,2x^2$$

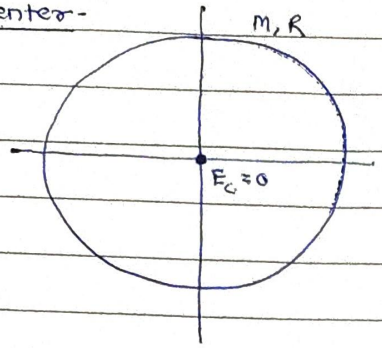
$$a^2+x^2 = 3x^2$$

$$2x^2 = a^2 \qquad \Rightarrow x = \pm \frac{a}{\sqrt{2}}$$

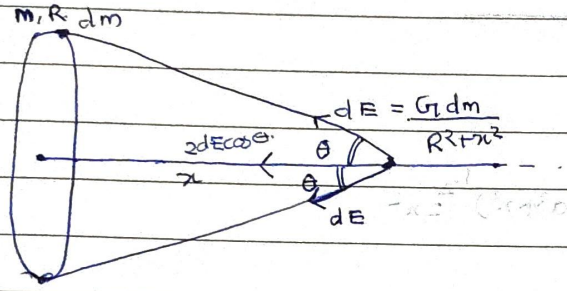


■ $\vec{E}_g$ due to uniform Ring :-

① at center-



② at axis :-



$$E_g = 2\int dE\cos\theta$$

$$= 2\int \frac{G_1 dm}{R^2+x^2} \cdot \frac{x}{\sqrt{R^2+x^2}}$$

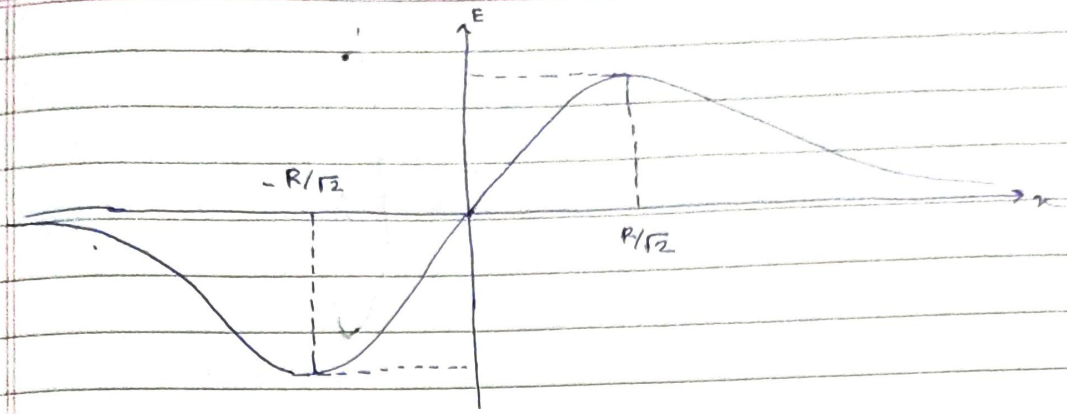
$$E_g = \frac{2Gx}{(R^2+x^2)^{3/2}} \int_0^{m/2} dm \quad = \frac{2Gx\ m/2}{(R^2+x^2)^{3/2}}$$

$$\boxed{E_g = \frac{GMx}{(R^2+x^2)^{3/2}}}$$

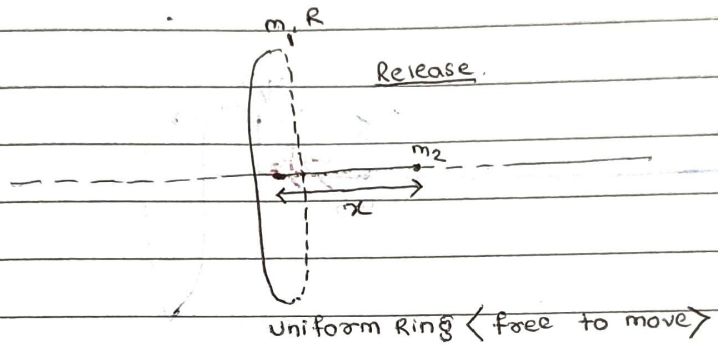
$x \to 0$ $\quad E_{near} = \frac{GMx}{R^3}$

$x \to$ large $\quad E_{far} = \frac{GM}{x^2}$ $\quad \langle R^2+x^2 = x^2 \rangle$

$E_c = 0.$

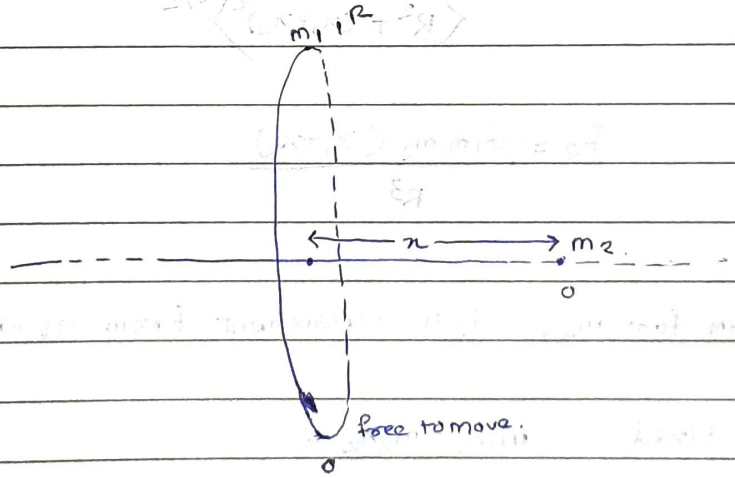


(q-5)



Find time period of their SHM.

ans-



$v_{com} = 0 = \text{const}$

$a_{com} = 0$

$x_{com} = 0$ $\xrightarrow{m_1 x_1}$ $\xleftarrow{m_2 x_2}$ $m_1 x_1 = m_2 x_2$.

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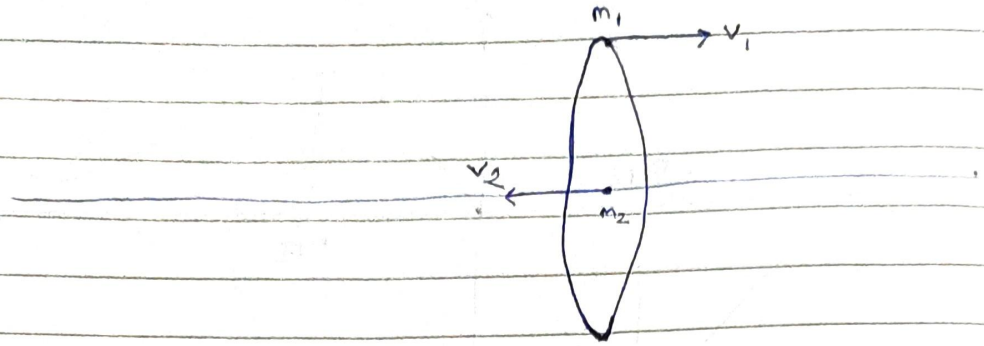
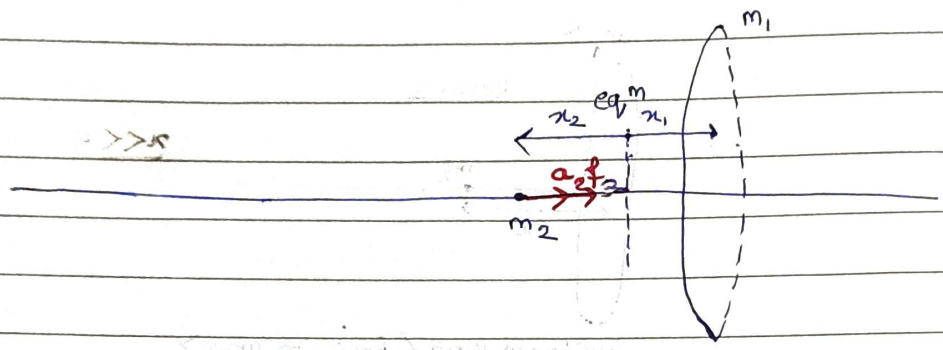


equilibrium.



$$f_2 = m_2 E_1$$

$$= m_2 \frac{G m_1 (x_1 + x_2)}{\left(R^2 + (x_1 + x_2)^2\right)^{3/2}}$$

$$f_2 = \frac{G m_1 m_2 (x_1 + x_2)}{R^3}$$

for SHM for m_2. $f_2 \propto$ displacement from equilibrium position.

How? com fixed $m_1 x_1 = m_2 x_2$

$$f_2 = \frac{G m_1 m_2 \left(\frac{m_2 x_2}{m_1} + x_2\right)}{R^3}$$

$$f_2 = \frac{G m_2 (m_1 + m_2) x_2}{R^3}.$$

Ans

$f_2 \propto x_2$ ⇒ SHM proved

$$f_2 = \frac{Gm_2(m_1+m_2)x_2}{R^3} = m_2 a_2 \qquad \because \langle f = ma \rangle$$

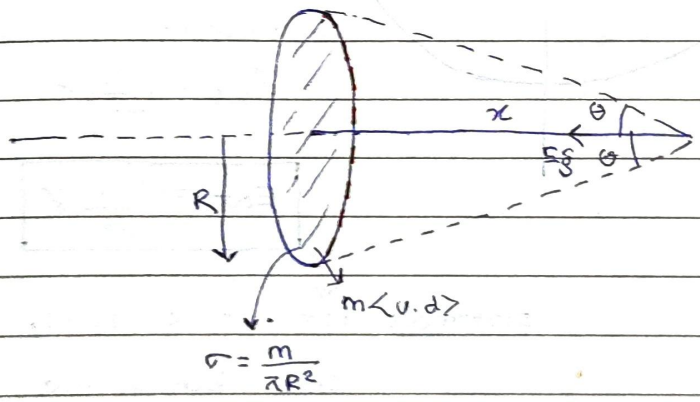
$$\Rightarrow a_2 = \frac{G(m_1+m_2)x_2}{R^3} \qquad \because \langle a = \omega^2 x \rangle$$

ω^2

$$\omega = \sqrt{\frac{G(m_1+m_2)}{R^3}} \quad \& \quad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R^3}{G(m_1+m_2)}}$$

Ans.

■ E_g due to uniform disc :-



$$E_g = \frac{\sigma}{2\varepsilon_0}\langle 1-\cos\theta \rangle$$

$\varepsilon_0 = \frac{1}{4\pi K}$

compare from electrostatic

$$E_g = \frac{\sigma 4\pi G}{2}\left\langle 1 - \frac{x}{\sqrt{R^2+x^2}} \right\rangle$$

Process :-

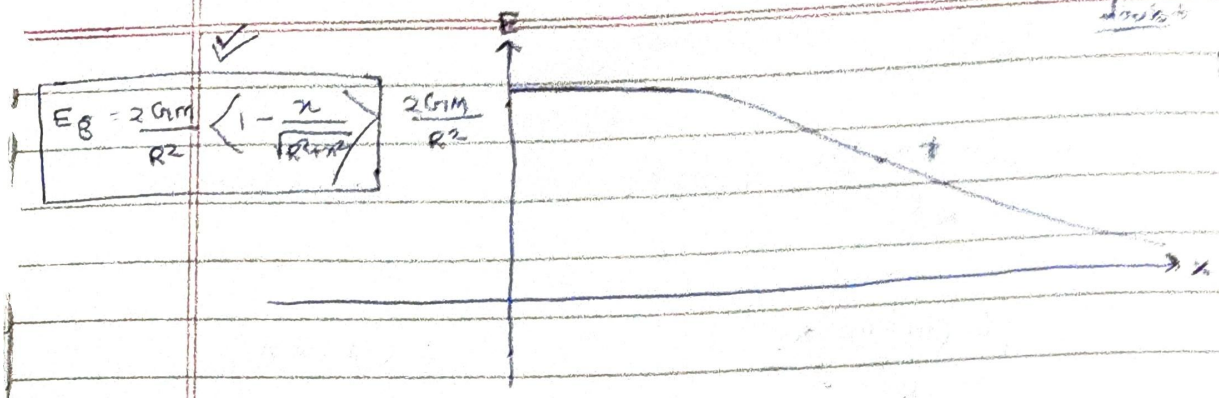
take small Ring strip & integrate it or match from electrostatic.

$$E_g = 2\pi G\sigma\left\langle 1 - \frac{x}{\sqrt{R^2+x^2}} \right\rangle$$

$$= \frac{2\pi GM}{\pi R^2}\left\langle 1 - \frac{x}{\sqrt{R^2+x^2}} \right\rangle$$

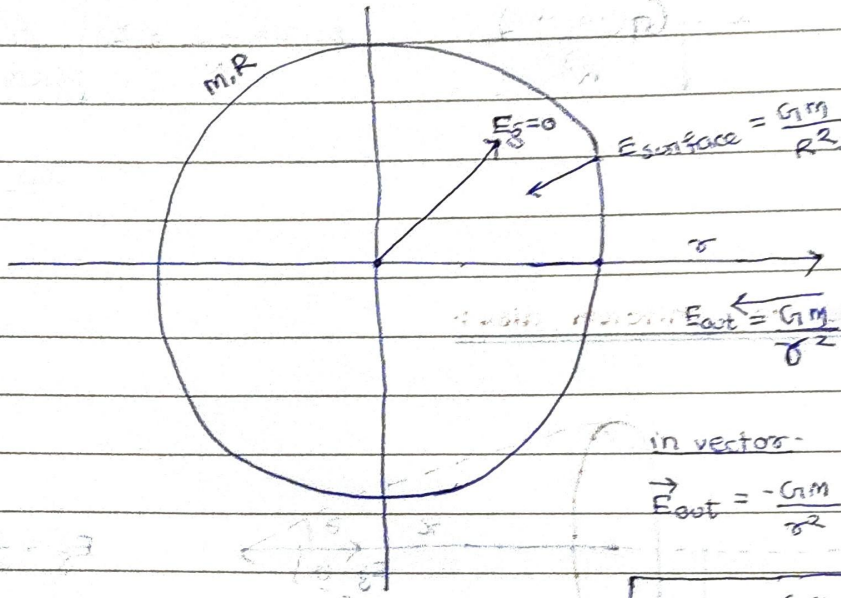
$$= \frac{2GM}{R^2}\left\langle 1 - \frac{x}{\sqrt{R^2+x^2}} \right\rangle$$

$$E_g = \frac{2GM}{R^2}\langle 1-\cos\theta \rangle$$



■ E_s due to Hollow uniform sphere:-

or spherical shell.

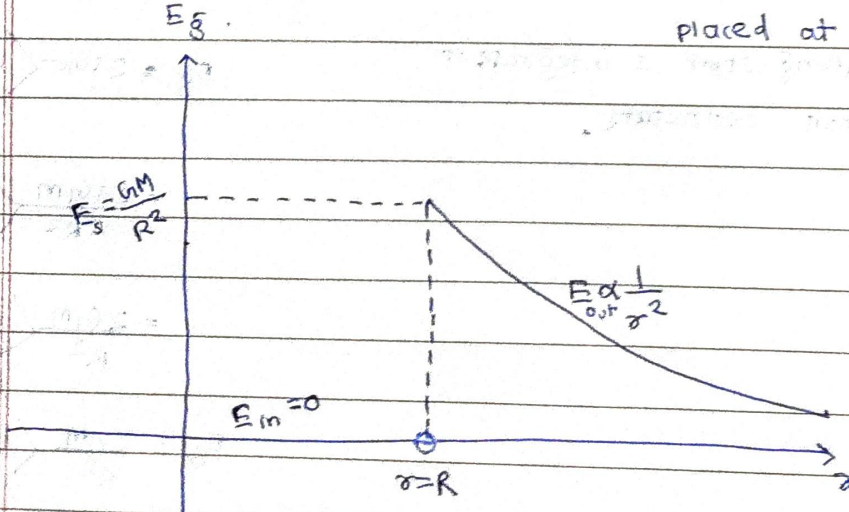


in vector-

$$\vec{E}_{out} = -\frac{GM}{r^2}\hat{r}$$

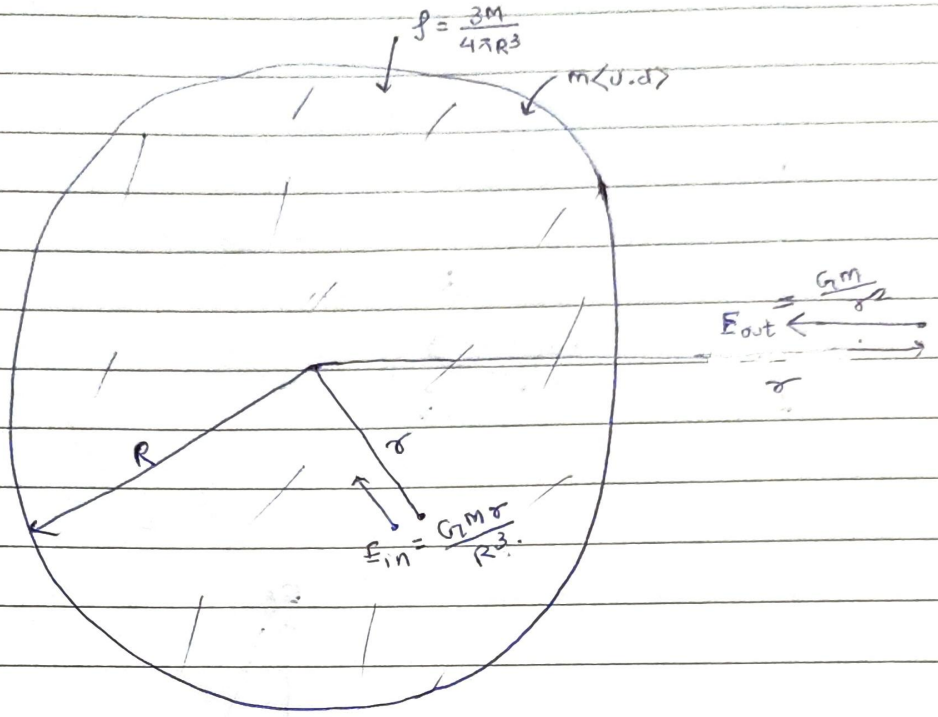
$$E_{out} = -\frac{GM}{r^3}\vec{r}$$

statement- for outside point, spherical shell behave like point mass placed at center.



#2

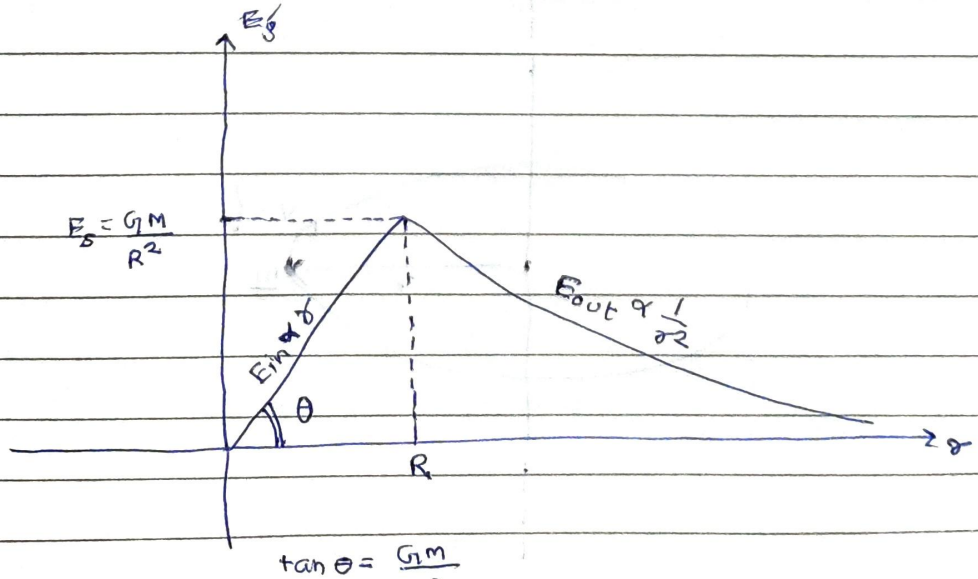
Gravitational field due to uniform solid sphere :- <imp>



in vector form

$$\vec{E_{in}} = \frac{GMr}{R^3}(-\hat{r}) = -\frac{GM\vec{r}}{R^3}$$

$$\vec{E_{out}} = \frac{GM}{r^2}(-\hat{r}) = -\frac{GM\vec{r}}{r^3}$$

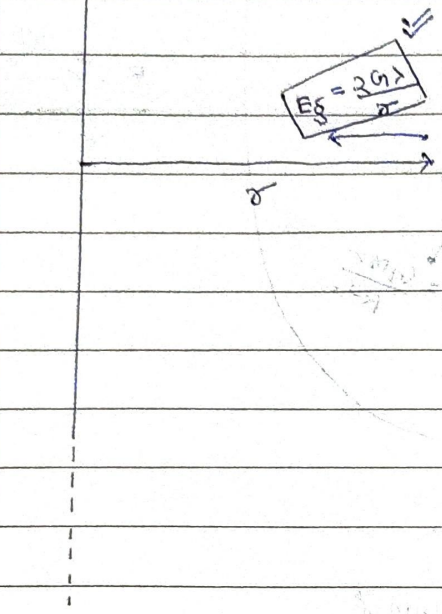


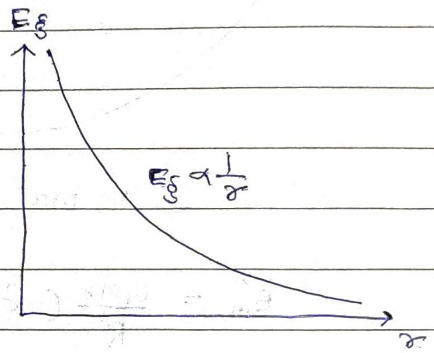
$$\tan\theta = \frac{GM}{R^3}$$

Gravitational field due to uniform long wire :

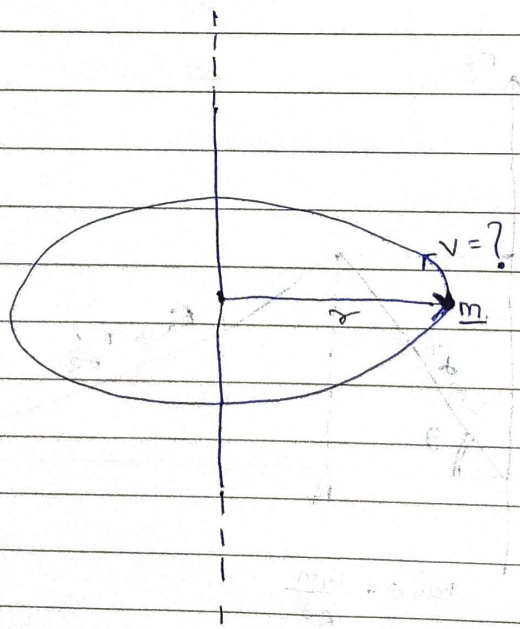
$\propto$ long

$\lambda = \frac{\text{mass}}{\text{length}}$

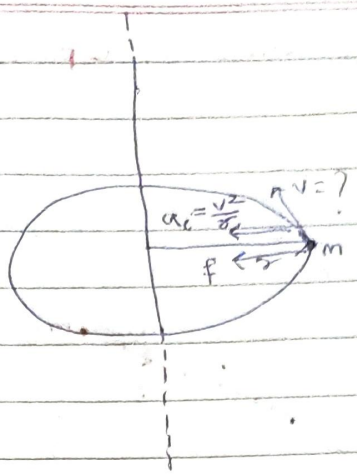




Q-6 find 'v' such that m performs UCM about infinity log wire.



Soln:-



Rem!

$a_c = \omega^2 r = \frac{v^2}{r}$

$f = mE_g$

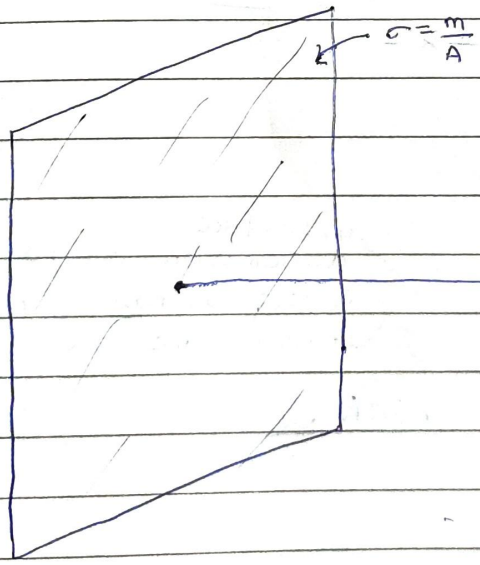
$f = m \times \frac{2GM}{r} = \frac{mv^2}{r}$

$\Rightarrow v^2 = 2GM$

$\Rightarrow v = \sqrt{2GM}$

$\because \omega = \frac{v}{r} = \sqrt{\frac{2GM}{r^2}}$ $\quad \because T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r^2}{2GM}}$

■ Gravitational field due to uniform infinitely large sheet:-



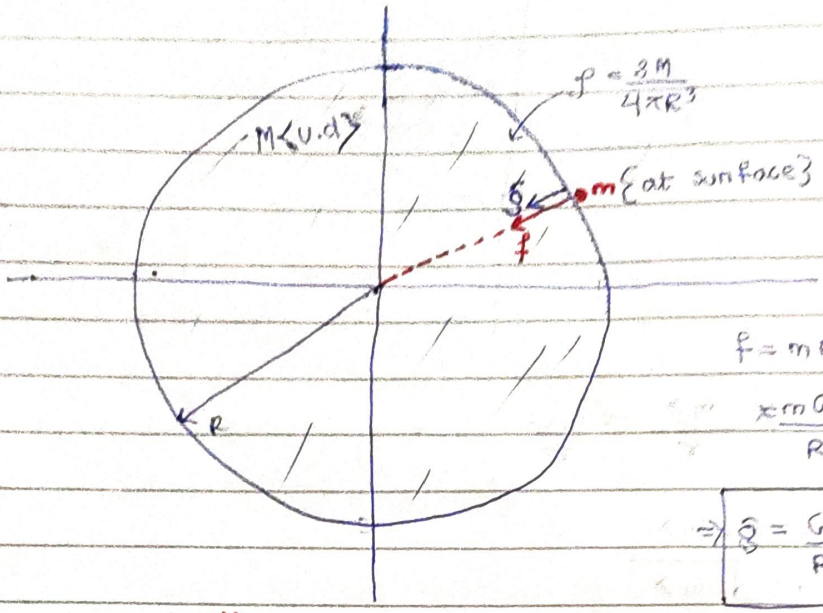
$E_g = \frac{\sigma \times 4\pi G}{2} = 2\pi G\sigma = E_g$.

in electrostatics $E = \frac{\sigma}{2\varepsilon_0}$

$E_g = 2\pi G\sigma$.

→ uniform.

Acceleration due to gravity on a planet :-



$f = mEg$

$\Rightarrow \frac{GmM}{R^2} = mg$

$\Rightarrow g = \frac{GM}{R^2}$ Rem.

at surface

#2

19.00 min

doubt.

JEE Advance concept.

Q- Two different planets are given, Radius (R) & mass (M) are different → obviously both diff properties. given. acceleration due to gravity given for 1st planet find acceleration due to gravity in 2nd planet.

H.W

— x — x — x —

$$g = \frac{GM}{R^2}$$

for 2 planets -

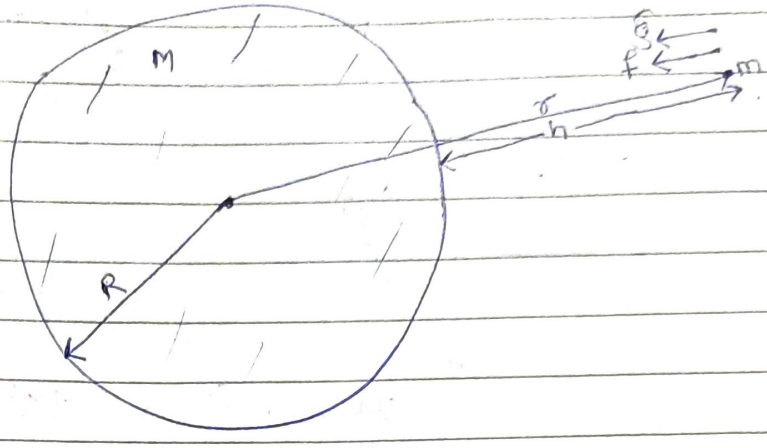
if same m, $g \propto \frac{1}{R^2}$

on surface acceleration due to gravity is less of bigger radius object having same mass

$$g = \frac{G\rho\frac{4}{3}\pi R^3}{R^2} = \frac{G\rho\pi R \times 4}{3}$$

if same ρ, $g \propto R$.

Variation of "g" with height :-



$$f = \left(\frac{GM}{r^2}\right) m. \quad = mg'$$

Recap: For outside point same behave like point mass placed at center.

$\left\langle f = m.E_g \right\rangle$ ans

$$\boxed{g' = \frac{GM}{r^2}}$$

in term of h. $\boxed{g' = \frac{GM}{(R+h)^2}}$ as $r = h+R$.

$h \uparrow \Rightarrow g' \downarrow$.

for near point.

$h << R \quad \left\{\frac{h}{R} << 1\right\}$

$$g' = \frac{GM}{R^2\left\langle 1+\frac{h}{R}\right\rangle^2}$$

$$g' = g\left\langle 1+\frac{h}{R}\right\rangle^{-2}$$

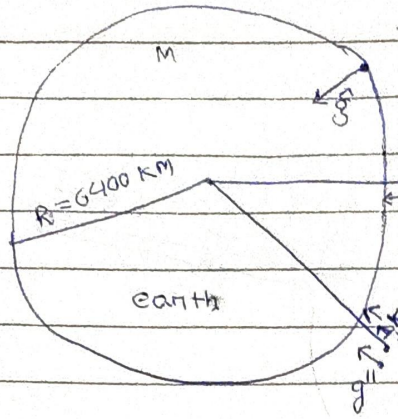
$$\boxed{g' = g\left\langle 1-\frac{2h}{R}\right\rangle}$$ only for near point.

using binomial approxim.

$(1+x)^n = 1+nx$

if $x << 1$.

Ex-



$g = \frac{GM}{R^2} = 9.8\ m/s^2$

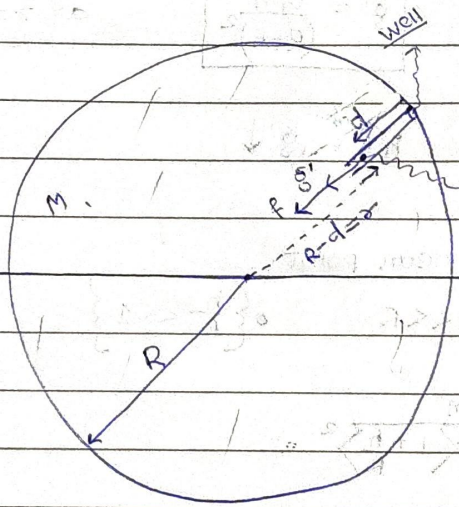
$g' = \frac{GM}{(2R)^2} = \frac{GM}{4R^2} = \frac{9.8}{4}\ m/s^2 = \frac{g}{4}\ m/s^2$

$g'' = g\left(1 - \frac{2h}{R}\right)$

ye bala formula only and only for near point. $= g\left(1 - \frac{2}{6400}\right) = g\left(\frac{6398}{6400}\right)$

Ye bala formula chahe najdeek ho chahe dur. you can always use it. we can find g'' by this formula answer more or less. same hi ata.

Variation of "g" with depth :-



well ka jo thickness hogi^jo (yani circumference hogi vo earth ke same negligible hogi.

→ particle having mass "m".

$f = \frac{m\,GM(R-d)}{R^3} = mg'$

$g' = \frac{GM}{R^2}\left(1 - \frac{d}{R}\right)$

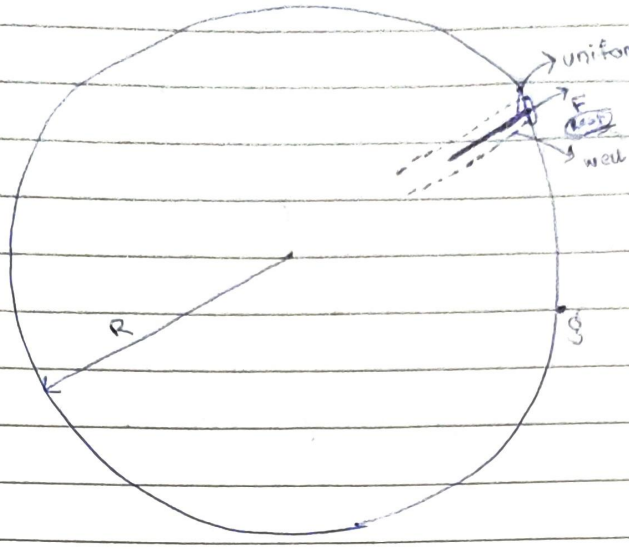
$\boxed{g' = g\left(1 - \frac{d}{R}\right)}$

at center $d = R$

$\Rightarrow g' = 0$

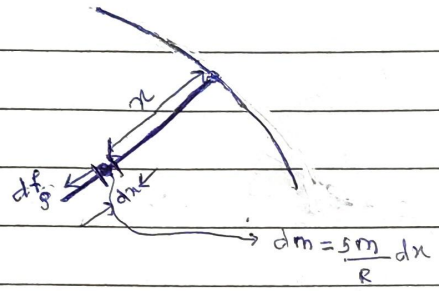
** accleration due to gravity is decreasing linierly with depth.

Q-3



find external force F so that rope remains at equilibrium.

Ans-



Rope ke har point pe g ka value same nehin he. So we integrate.

$$df_g = dm\,g'$$

$$= \frac{5m\,dx}{R}\,g\left\langle 1 - \frac{x}{R}\right\rangle$$

$$f_g = \int_0^{R/5} \frac{5m}{R}\,dx\,g\left\langle 1-\frac{x}{R}\right\rangle = \frac{5mg}{R}\int_0^{R/5}\left(1-\frac{x}{R}\right)dx$$

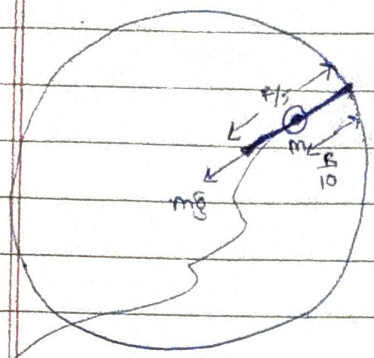
$$f_g = \frac{5mg}{R}\left\{x - \frac{x^2}{2R}\right\}_0^{R/5} = \frac{5mg}{R}\left[\frac{R}{5} - \frac{R^2}{25\times 2R}\right]$$

$$= mg\left[1 - \frac{1}{10}\right] = \frac{9mg}{10}$$ Ans

for rope to be in equilibrium. we apply $F_{ext} = F_g$.

so, $\boxed{F_{ext} = \frac{9mg}{10}}$ which is opposite to $\vec{F}_g$ — (i).

Trick- only work when function varies linearly.



$$f_g = mg\left\langle 1 - \frac{R}{10R} \right\rangle$$

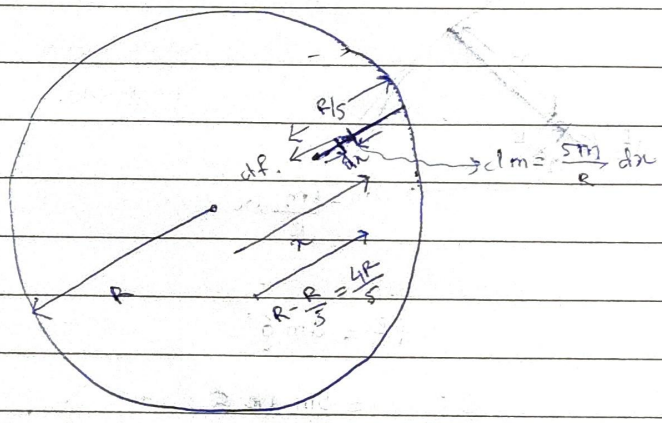
$$= \frac{9mg}{10}$$

same as (i)

near 'g' varry linearly with x. so we can take Average.

Assume total mass is at this point. math allow to do that.

Alitar-1



$$df_g = dm\left(\frac{GM\,x}{R^3}\right) \qquad \left\langle f = \frac{GMm}{R^2} \right\rangle \; \therefore F_{in} = \frac{Gmx}{R^3}$$

$$f_g = \int df_g = \int \left(\frac{5m}{R}dx\right)\frac{GM}{R^3}x$$

$$= \left(\frac{5m}{R}\right)g \times \frac{1}{R}\int_{\frac{4R}{5}}^{R} x\,dx$$

$$= \frac{5gm}{R^2}\left[\frac{x^2}{2}\right]_{\frac{4R}{5}}^{R} \quad \therefore \frac{5gm}{R^2}\left\langle \frac{R^2}{2} - \frac{16R^2}{25\times 2} \right\rangle = \frac{5gm}{R^2}\left\langle \frac{25R^2 - 16R^2}{50} \right\rangle$$

* A rotating frame of reference is a special case of a non-inertial reference frame that is rotating relative to an inertial reference frame.

* In rotating frame of reference same ke same force ke sath ham ek extra force show karte he that is 'pseudo force' mean it is centrifugal force.

$$= \frac{5}{R^2}\frac{gm}{}\left\langle \frac{qR^2}{5q_c} \right\rangle$$

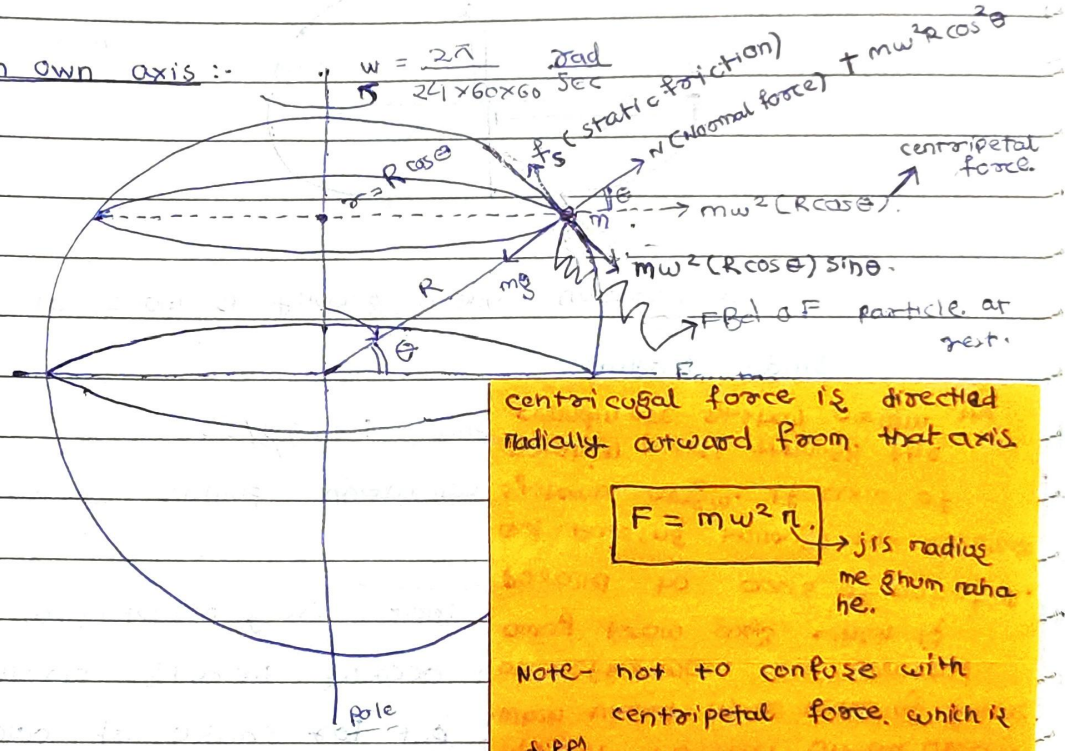
$$f_s = \frac{qgm}{10}$$

same as (i)

$$\Rightarrow F_{ext} = \frac{qgm}{10}$$ Ans

Variation of g with rotation of earth :-

rotation on own axis :-



centrifugal force is directed radially outward from that axis

$$F = m\omega^2 r.$$ → jis radius me ghum raha he.

Note- not to confuse with centripetal force which is diff.

θ → lattitude

90-θ → co-lattitude.

$$f_s = m\omega^2 R\cos\theta\sin\theta.$$

$$f_s = \frac{m\omega^2 R\sin2\theta}{2}$$

$$mg = N + m\omega^2 R\cos^2\theta$$

$$N = mg - m\omega^2 R\cos^2\theta = W_{apparent}.$$

$$= mg_{eff}.$$

$$\boxed{g_{eff} = g - \omega^2 R\cos^2\theta}$$

$$g_{eff} = g - \omega^2 R \cos^2\theta$$

* g is max at pole.

Pole: $\theta = 90^\circ$

$N = W_{app} = mg$

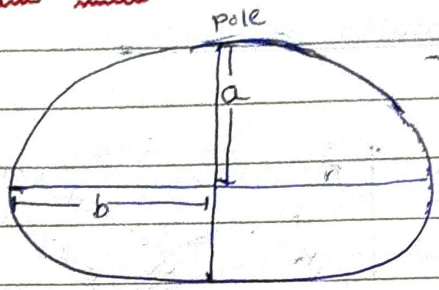
$g_{eff} = g$

equator: $\theta = 0^\circ$

$N = W_{app} = mg - m\omega^2 R$

$g_{eff} = g - \omega^2 R$

■ **interesting facts :-** shape of earth is not exact sphere



b > a.

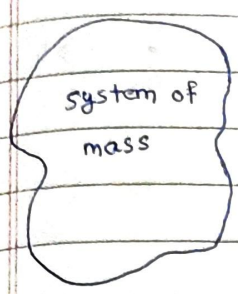
so acceleration due to gravity is more at pole, bcz of small radius.

The diff^ is approx 1.82 cm/s².

for shake of calculation purpose we assume it as perfect sphere.

It is not necessary that every planet is a perfect sphere. If small deformation occurs there is small change in g acc^ due to gravity, but for shake of calculation we assume them perfect sphere.

Gravitational potential :-



(test mass)

for def^n refer electrostatics.

$$V_p = \frac{(W_{ext})_{\infty \to P}}{m_0} \quad \text{(slowly)} = \frac{-(W_{\vec{g}})_{\infty \to P}}{m_0} = \frac{(W_{\vec{g}})_{P \to \infty}}{m_0}$$

$$= \frac{-\int_{\infty}^{P} m_0 \cdot \vec{E}_{\vec{g}} \cdot d\vec{r}}{m_0}$$

$$V_p = -\int_{\infty}^{P} \vec{E}_{\vec{g}} \cdot d\vec{r}$$

$$V_p = \int_{P}^{\infty} \vec{E}_{\vec{g}} \cdot d\vec{r} \rightarrow \text{Line integral of gravitational field.}$$

- unit : $\frac{\text{Joule}}{\text{Kg}}$
- scalar <bcz Joule & Kg are scalar>.
- $V_\infty = 0$
- dim: $[L^2 T^{-2}]$

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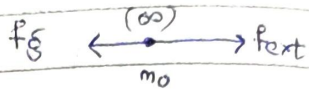
● Gravitational potential due to point mass :- (or (V.))



W_g = +ve.

W_{ext} = -ve.

$$V = \frac{W_{ext}}{m_0}$$

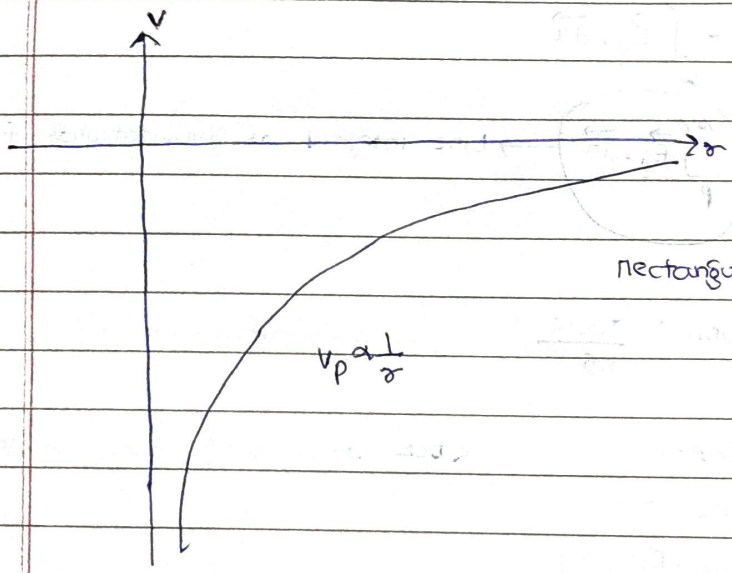
$$= \frac{-W_g}{m_0}$$

$$V_p = \frac{-GM}{r}$$

see process in electrost

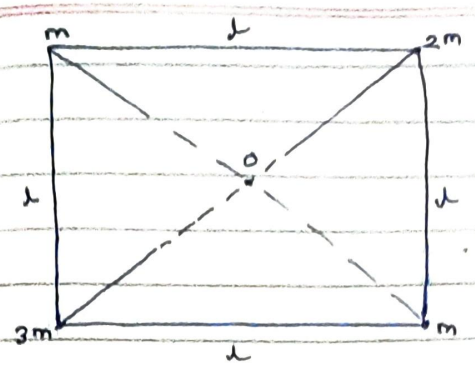
$$V = \frac{KQ}{r}$$

$K \to -G$.



rectangular hyperbola

Q1- B



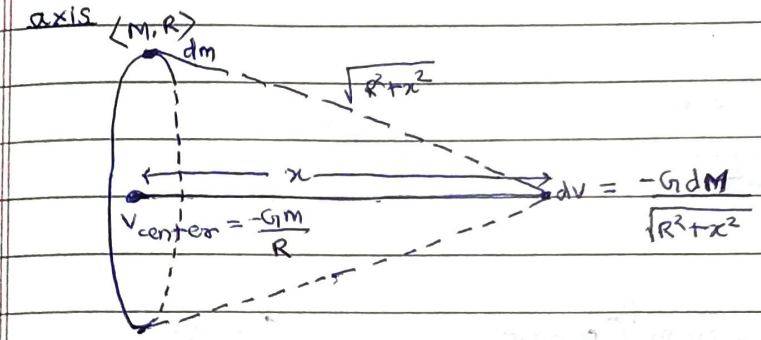
Find V_O (Gravitational Potential).

Sol:- $MO = \frac{\sqrt{2}l}{2} = \frac{l}{\sqrt{2}}$

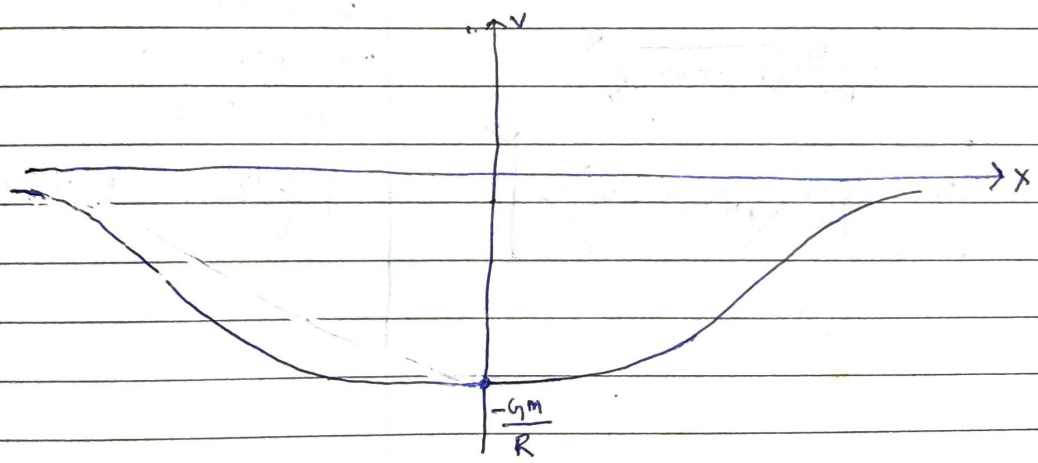
$$V_O = \frac{-Gm}{l/\sqrt{2}} - \frac{G(2m)}{l/\sqrt{2}} - \frac{G(m)}{l/\sqrt{2}} - \frac{G(3m)}{l/\sqrt{2}}$$

$$= \frac{-G}{l/\sqrt{2}}(7m) = \frac{-7\sqrt{2}\,Gm}{l}$$

- **V. due to ring :-**

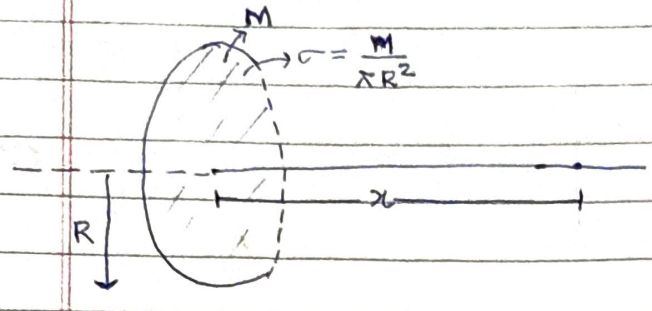


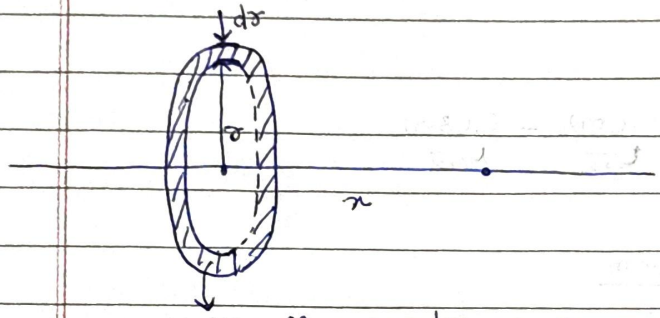
$$V = \int dV = \frac{-G}{\sqrt{R^2+x^2}} \int dM = \frac{-GM}{\sqrt{R^2+x^2}}$$



NOTE - in potential we dont care about mass distribution, means uniform or not.

V due to uniform disc :-





$dM = \frac{M}{\pi R^2} \times 2\pi r dr = \frac{M}{R^2} 2r dr.$

So, $dv = \frac{-G dM}{\sqrt{r^2+x^2}}$

$= -G \frac{M}{R^2} \frac{2r dr}{\sqrt{r^2+x^2}}$ <using ?>

$V = \int dv = \frac{-2GM}{R^2} \int_0^R \frac{r dr}{\sqrt{r^2+x^2}}$

$\int \frac{r dr}{\sqrt{r^2+x^2}}$

$r^2+x^2 = t^2$

$2r dr + 0 = 2t dt.$

$r dr = t dt.$

$\int \frac{t dt}{t} = t = \sqrt{r^2+x^2}$

$= \frac{-2GM}{R^2} \left(\sqrt{r^2+x^2} \right)_0^R$

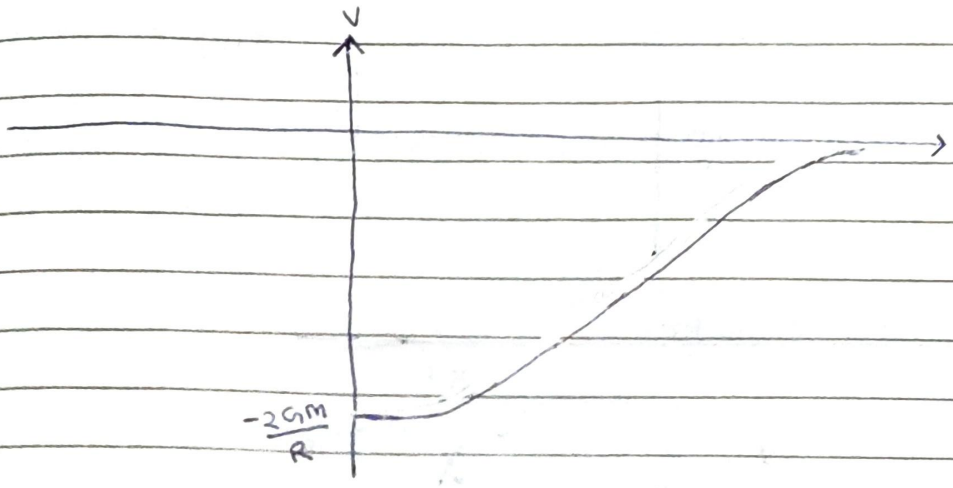
$V = \frac{-2GM}{R^2} \left\langle \sqrt{R^2+x^2} - x \right\rangle$ Ans!

by rationalise

$$V = \frac{-2GM}{R^2}\left\langle \sqrt{R^2+x^2} - x \right\rangle \left\langle \frac{\sqrt{R^2+x^2}+x}{\sqrt{R^2+x^2}+x} \right\rangle$$

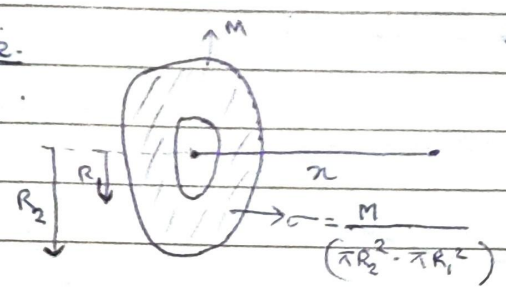
$$= \frac{-2GM}{R^2} \frac{R^2}{\sqrt{R^2+x^2}+x}$$

$$\boxed{V = \frac{-2GM}{\left(\sqrt{R^2+x^2}+x\right)}}$$ Rem!



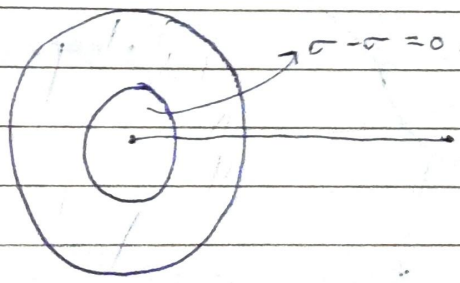
i.e.

method-1



For answer take limit, from R_1 to R_2.

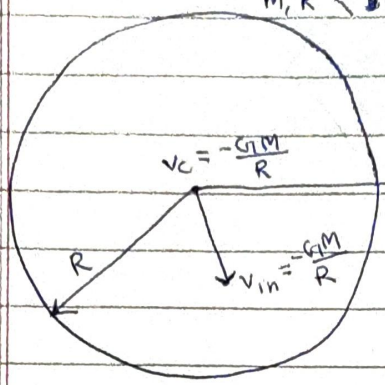
method-2

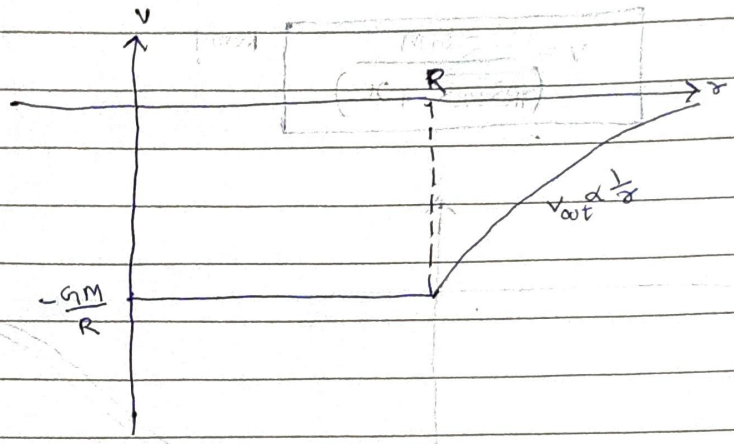


#3

• V due to spherical shell :-

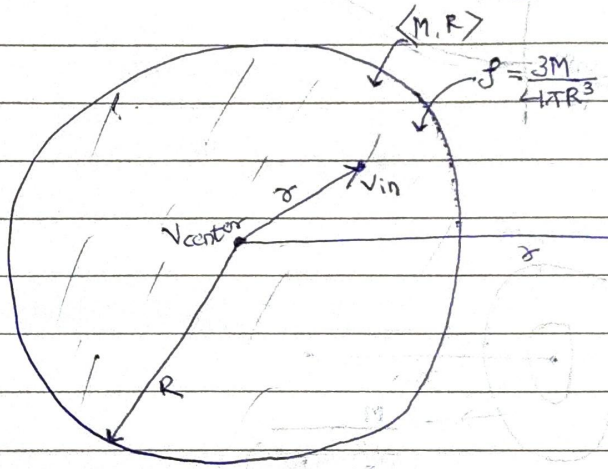
M, R ⟨Hollow shell⟩





• V due to uniform Solid sphere :-

imp

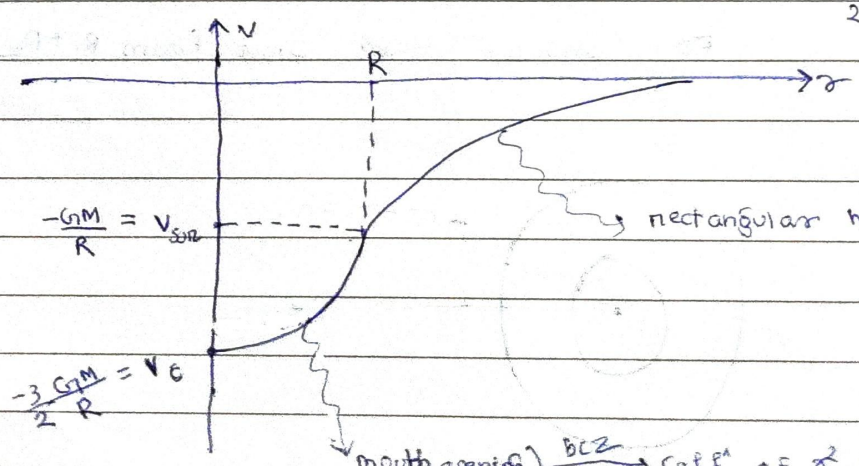


$V_{out} = \frac{-GM}{r}$

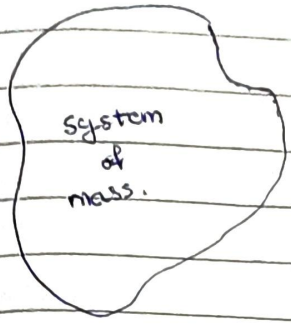
$V_{in} = \frac{-GM}{2R^3} \langle 3R^2 - r^2 \rangle$

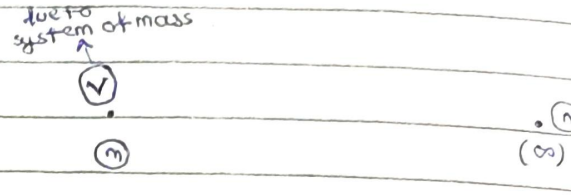
for $r = 0$.

$V_{center} = \frac{-3GM}{2R}$



• Gravitational potential energy :- V





$$U = Vm.$$

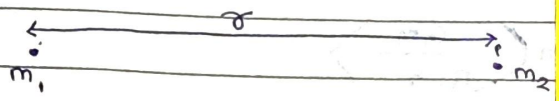
1.00 min

doubt

Just note

Work done is change in total mechanical energy

• G.P.E. of 2 point masses :-



$$W_{ext} = V_1 m_2 = U$$

$$= -\frac{GM_1}{r} m_2 = U$$

in detail :-

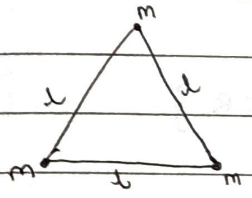
First m_1, m_2 ko ∞ bhej do. then m_1 ko slowly lao there is no force to attract or repeal m_1, so ap negligible force se m_1 ko ∞ se laoge so work done is zero.

Now if we m_2 ko ∞ se layenge hamko koina na koi work done karna to padega.

— × — × —

1kg mass ko ∞ se p rake Lane me work done karnapadega, V_1 by m_1 ↑ to m_2 mass kelive. $W_{ext} = m_2 V_1$

Q-9

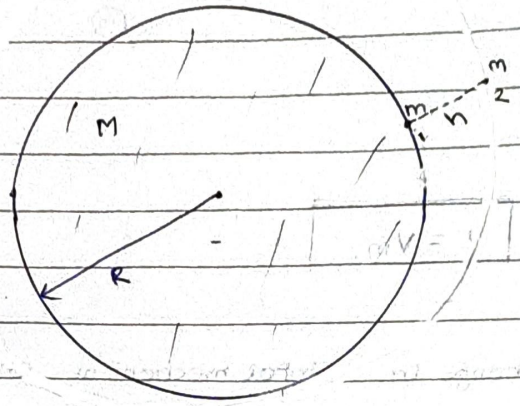


Find G.P.E of system.

Soln- $\frac{-Gm^2}{l} \times 3.$

$$\text{No. of terms} = {}^{n}C_2 = \frac{n(n-1)}{2}$$

Gravitational Potential energy near surface of planet :-



$h << R$

$$U_1 = m\left(\frac{-GM}{R}\right)$$

$$U_2 = m\left(\frac{-GM}{R+h}\right)$$

$U_2 - U_1$ = change in P.E.

$$= GMm\left(\frac{1}{R} - \frac{1}{R+h}\right)$$

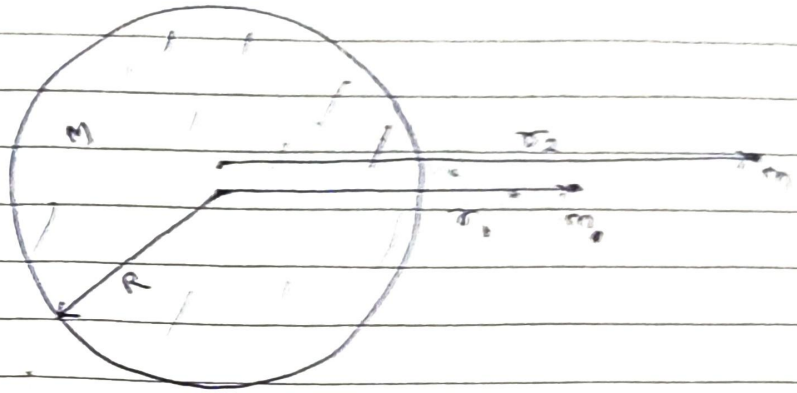
$$= \frac{GMm\ h}{R(R+h)}$$

$$= \frac{GMmh}{R^2} \qquad \{R+h \approx R\}$$

$$\boxed{\Delta U = mgh}$$

For $h << R$ means small dist^n.

Gravitational energy for a planet :-



$$U_i = m\left\langle \frac{-GM}{r_1} \right\rangle$$

$$U_f = m\left\langle \frac{-GM}{r_2} \right\rangle$$

$$\Delta U = U_f - U_i = GMm\left\langle \frac{1}{r_1} - \frac{1}{r_2} \right\rangle$$

in general, if dist^n is large.

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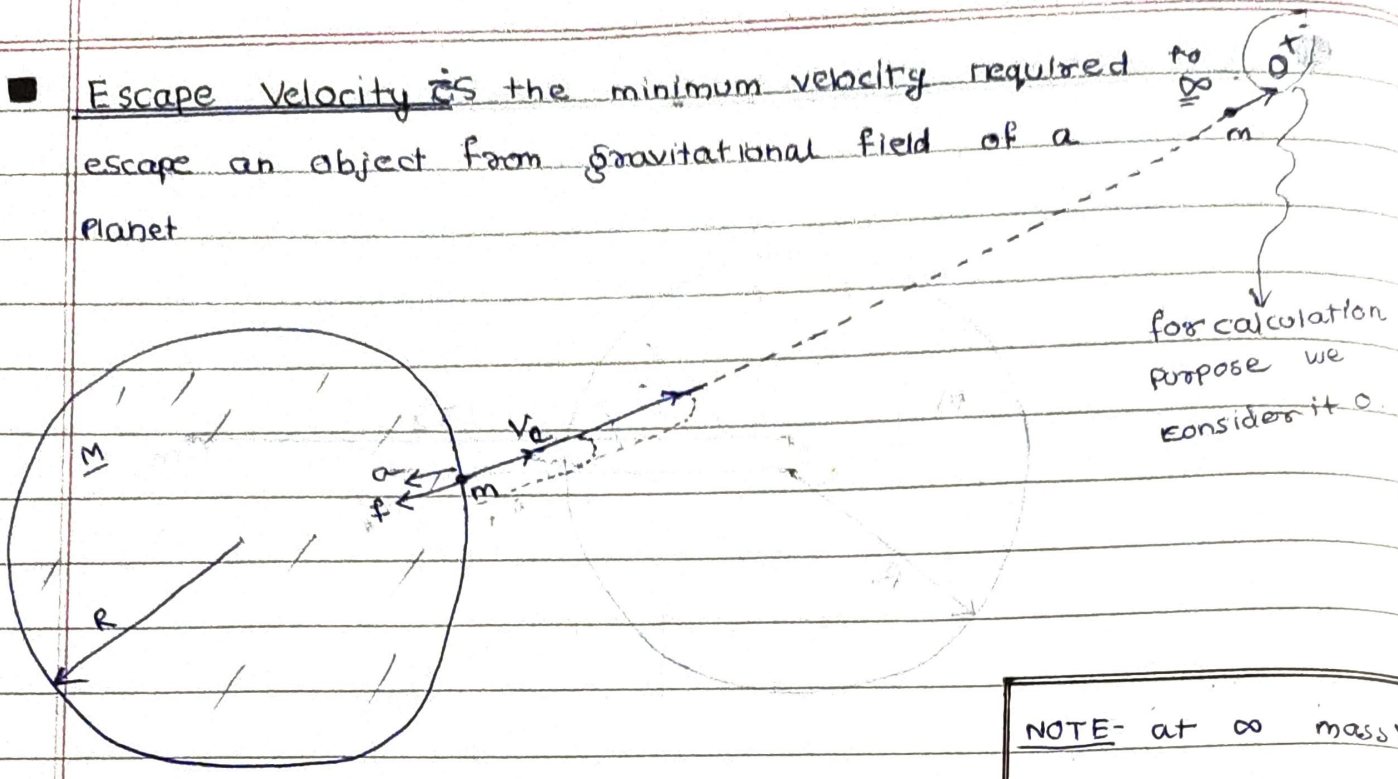
- **Escape Velocity** is the minimum velocity required to escape an object from gravitational field of a planet



TMEC

$$K_i + U_i = K_f + U_f$$

$$\frac{1}{2}mv_e^2 + m\left(-\frac{GM}{R}\right) = 0 + m\left(-\frac{GM}{\infty}\right)$$

$$\boxed{V_e = \sqrt{\frac{2GM}{R}} = \sqrt{2\left(\frac{GM}{R^2}\right)R}}$$

$$\boxed{V_e = \sqrt{2gR}}$$

NOTE- at ∞ mass 'm' planet ke gravi^ field ke bahar he, and planet ab koi force nehin Lagu rahahe. and it will not return to ~~earth~~ planet ke g. field and escape.

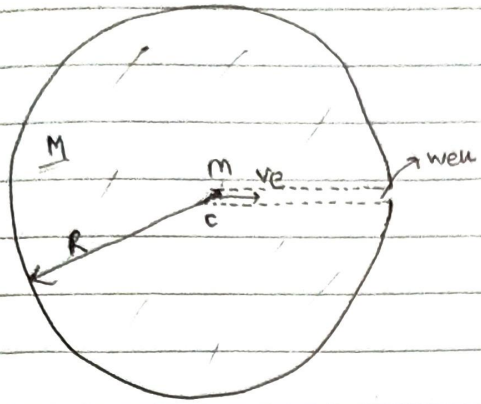
at ∞ → There is no. g. field of planet. to mass 'm' ko ane me koi sabal nehi utha.

for earth - $V_e = \sqrt{2 \times 9.8 \times 6400{,}000}$

$= 11.2 \times 10^3$ m/s

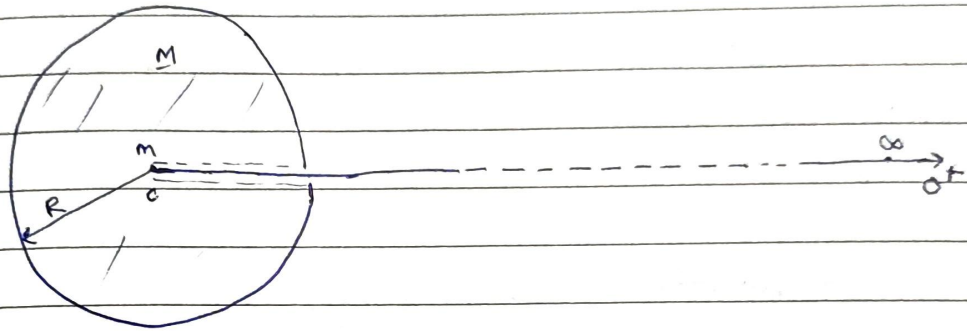
$= 11.2$ km/sec.

Q-10



find ve.

Soln -



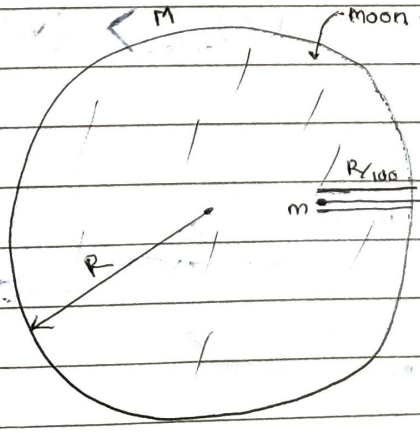
TMEC -

$K_i + U_i = K_f + U_f$

$\frac{1}{2}mv_e^2 + m\left(\frac{-3GM}{2R}\right) = 0 + 0$

$v_e = \sqrt{\frac{3GM}{R}}$

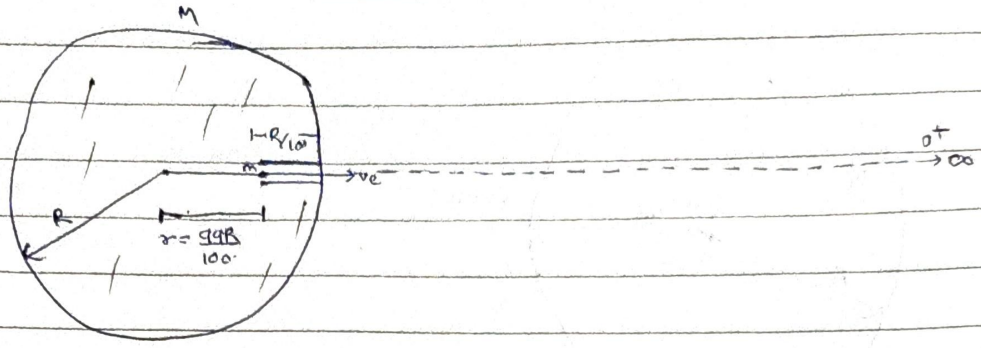
Q-11



Find v_e.

Create - Ojha / Gaddha.

Soln-



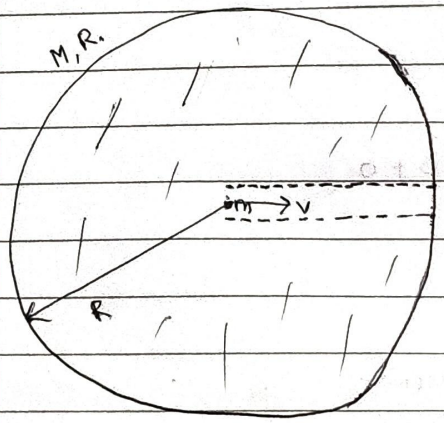
TMEC.

$$\frac{1}{2}mv_e^2 + m\left\langle \frac{-GM}{2R^2}\left(3R^2 - \left(\frac{99R}{100}\right)^2\right)\right\rangle = 0 + 0.$$

IIt bale sale chotiye hain sabal easy dekne kai liye value ma 6. mathete hain.

Solve yourself.

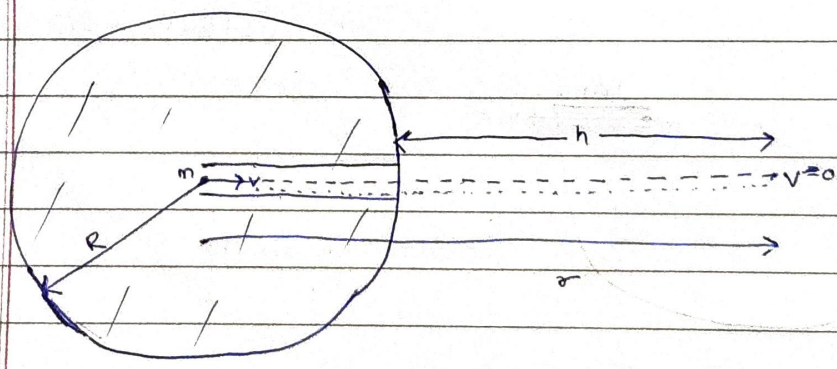
Q-12



v_i is equal to escape velocity from surface then find then find maximum height achieved by Particle (m) from surface of planet.

$$(v_e)_{center} > (v_e)_{surf}$$

Soln-



TMEC: $K_i + U_i = K_f + U_f$

$$\frac{1}{2}m\left(\frac{3GM}{R}\right) + m\left(\frac{-3}{2}\frac{GM}{R}\right) = 0 + m\left\langle\frac{-GM}{r}\right\rangle$$

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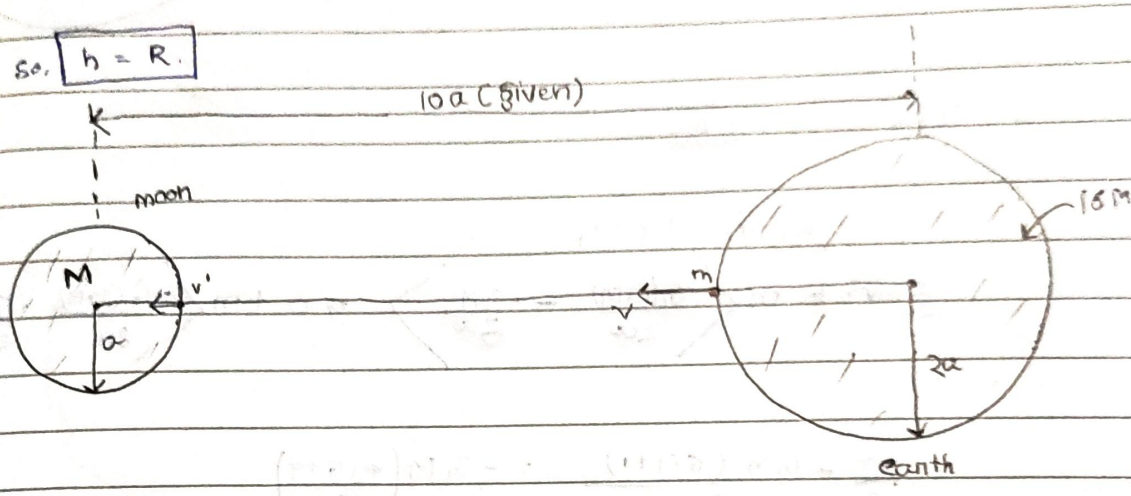
$$\frac{1}{R} - \frac{3}{2R} = \frac{-1}{r}$$

$$\frac{2-3}{2R} = \frac{-1}{2R} = \frac{-1}{r}$$

$$\Rightarrow r = 2R$$

So, $\boxed{h = R.}$

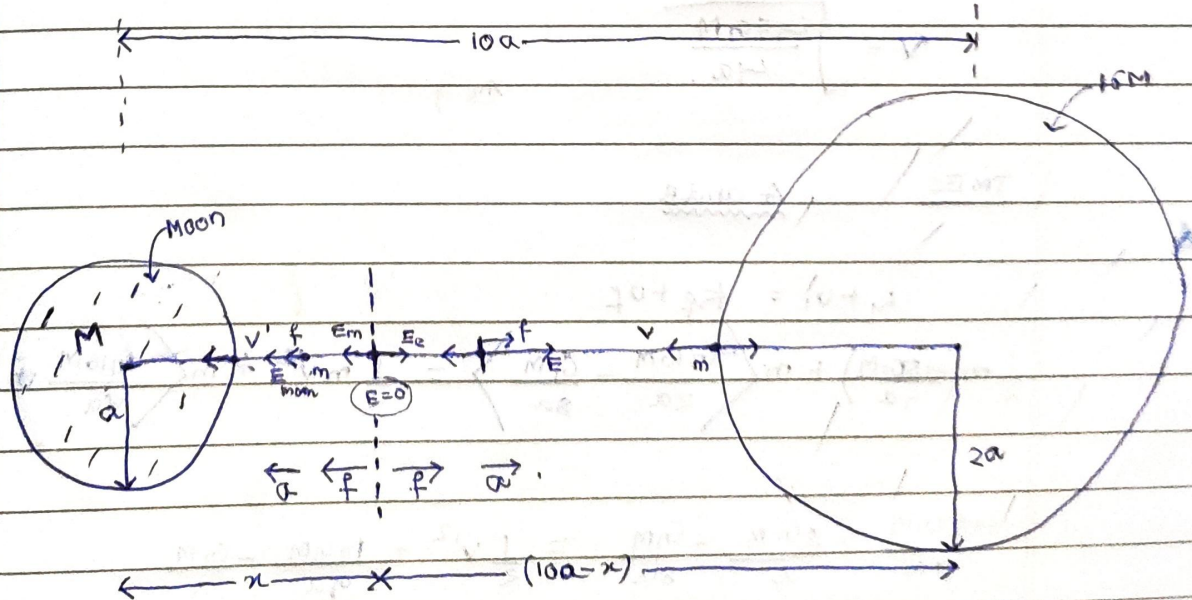
Q-13



(i) find minimum value of v so that particle can land on moon. (v = ?)

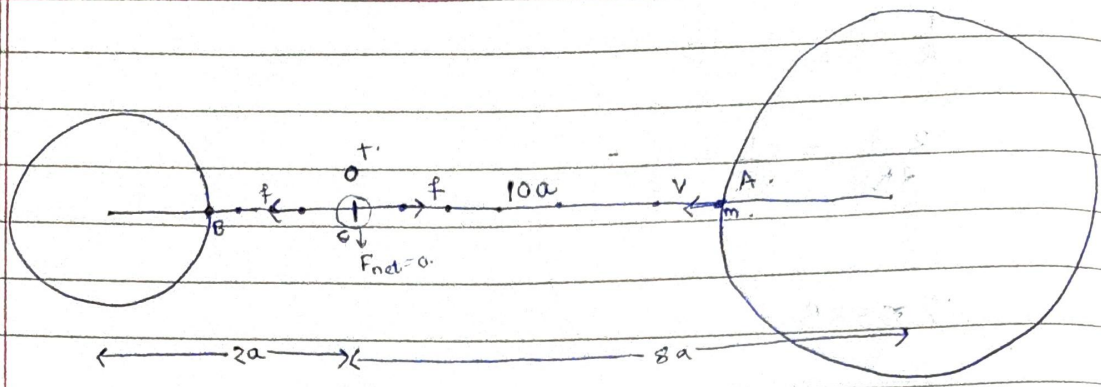
(ii) find the speed of particle when it lands on the moon. (v' = ?)

ans:-



$$\frac{GM}{x^2} \quad E=0 \quad \frac{G\,16M}{(10a-x)^2}$$

$$\Rightarrow 16x^2 = (10a-x)^2 \quad \Rightarrow 4x = 10a - x \quad \Rightarrow x = 2a$$



TMEC ABC

$K_i + U_i = K_f + U_f$

$$\frac{1}{2}mv^2 + m\left\langle \frac{-G(16M)}{2a} - \frac{GM}{8a} \right\rangle = 0 + m\left\langle \frac{-G(16M)}{8a} - \frac{GM}{2a} \right\rangle$$

$$\frac{v^2}{2} - \frac{GM(64+1)}{8a} = -GM\left(\frac{16+4}{8a}\right)$$

$$\frac{v^2}{2} = \frac{GM}{8a}\langle 65 - 20 \rangle$$

$$v = \sqrt{\frac{45GM}{4a}}$$

Ans

TM EC A and B

$K_i + U_i = K_f + U_f$

$$\frac{1}{2}m\left(\frac{45GM}{4a}\right) + m\left\langle \frac{-G16M}{2a} - \frac{GM}{8a} \right\rangle = \frac{1}{2}mv'^2 + m\left\langle \frac{-G16M}{9a} - \frac{GM}{a} \right\rangle$$

$$\frac{45GM}{8a} - \frac{8GM}{a} - \frac{GM}{8a} = \frac{1}{2}v'^2 - \frac{16GM}{9a} - \frac{GM}{a}$$

$$v'^2 = \frac{2GM}{a}\left\langle \frac{45}{8} - 8 - \frac{1}{8} + \frac{16}{9} + 1 \right\rangle$$

$$v'^2 = \frac{2GM}{a}\left\langle \frac{405 - 504 - 9 + 128}{72} \right\rangle$$

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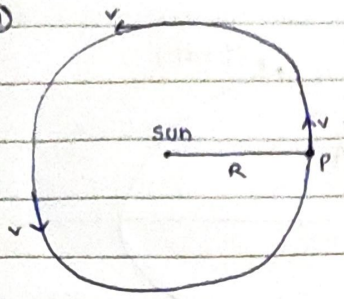
$$V^{1R} = \frac{GM}{a} \quad \frac{5-33-813}{36} = \frac{GM}{a} \quad \frac{20}{36} = \frac{5GM}{9a}.$$

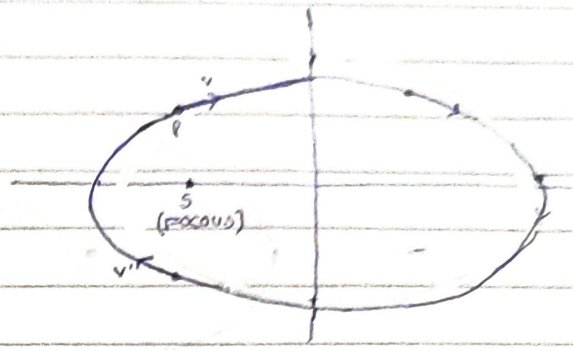
$$V^{1R} = \sqrt{\frac{5GM}{9a}}$$

Ans.

Planetary motion :-

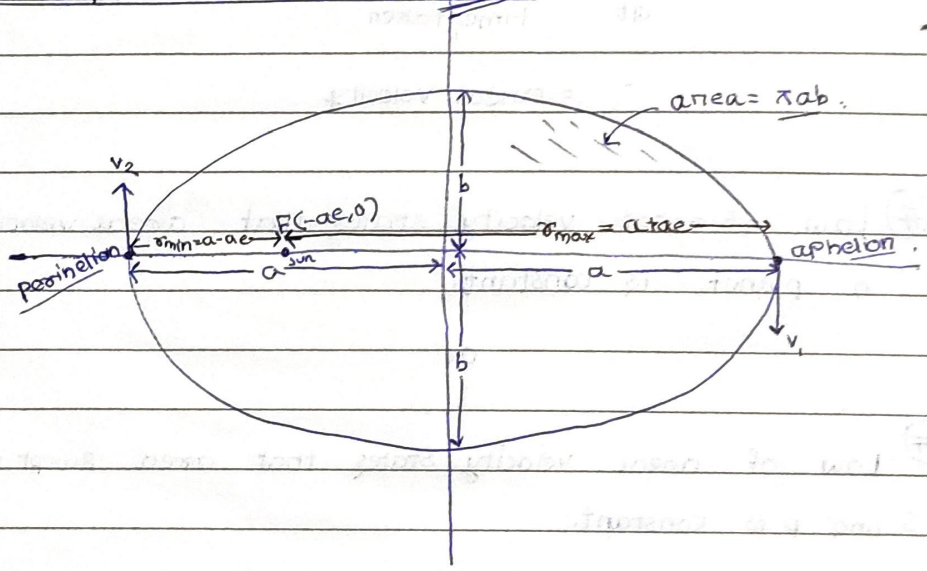
Kepler's law :-

① 

or 

it states that planet moves along a circle having sun as center or planet moves along an elliptical path having sun at one of the focus.

- Some important fact about ellipse :-



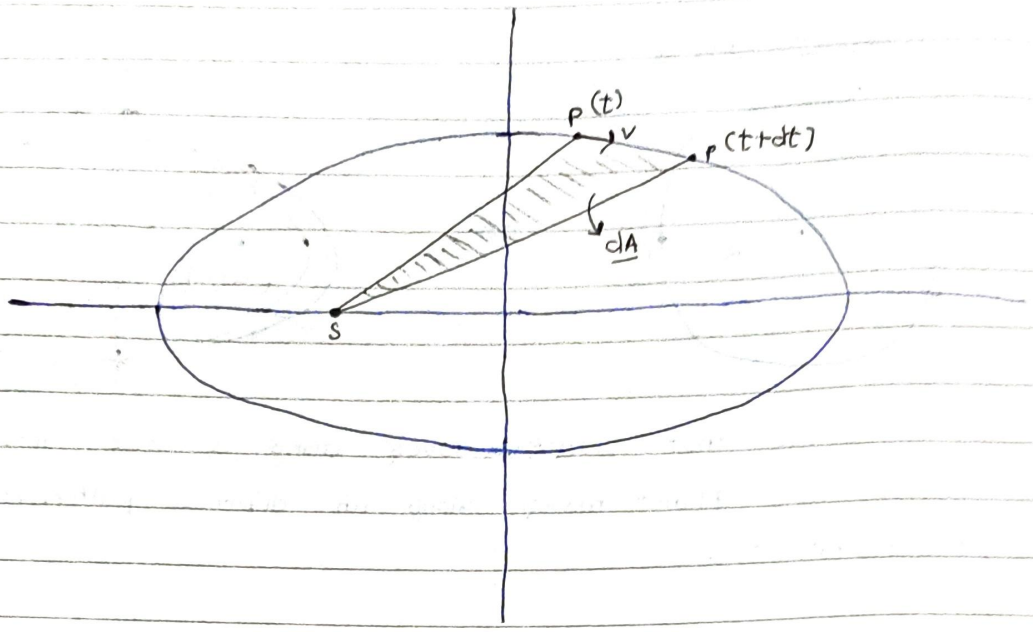
a - semi-major axis.

b - semi-minor axis.

e - eccentricity.

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

② Law of areal velocity :-



$\frac{dA}{dt}$ = $\frac{\text{area swept}}{\text{time taken}}$ unit : m^2/sec

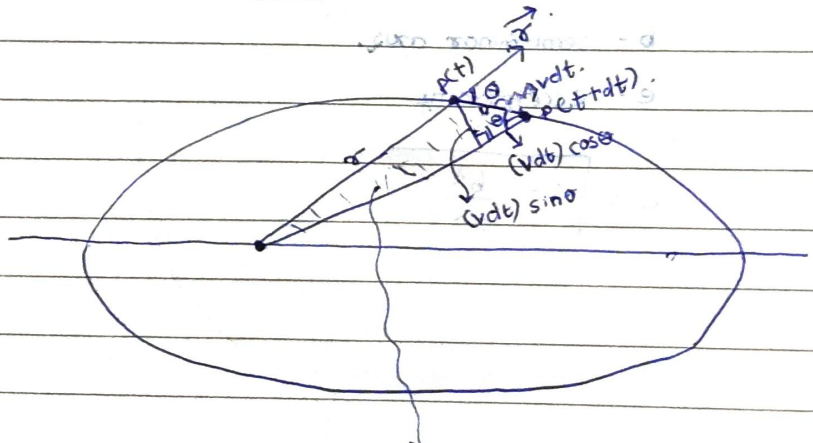
= areal velocity

Law of areal velocity states that areal velocity for a planet is constant.

or

Law of areal velocity states that area swept of line joining S and P is constant.

Proof :-



$dA = \frac{1}{2} \times r \times v\,dt \sin\theta$ (approx.)

$\vec{L} = m(\vec{r} \times \vec{v})$ classmate

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Areal velocity $= \frac{dA}{dt} = \frac{rv\sin\theta}{2}$

Planet per ek Lota he force kis naha he by sun.
so, sun ke about planet per jo tourque ayega is zero ($\tau = 0$).

$\Rightarrow$ tonque on planet about sun $\tau = 0$.

jis vi axis ya point ke about $\tau = 0$ hota he iske about angular momentum conserve hota he.

so, we can apply angular momentum conservation theorem.

$L = mvr\sin\theta$

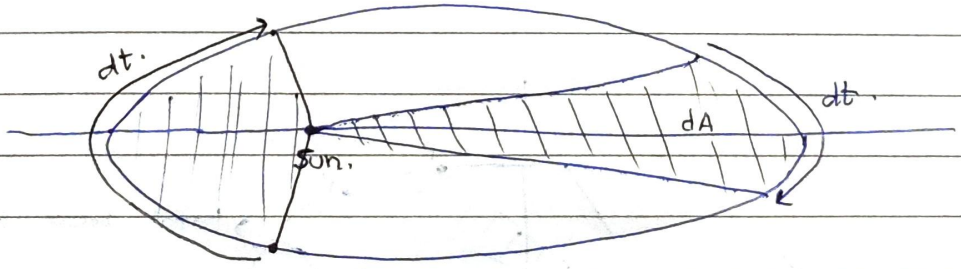
$\Rightarrow \frac{L}{m} = vr\sin\theta = \text{const.}$

constant $\rightarrow$ bcz $\Rightarrow \tau = 0$

constant.

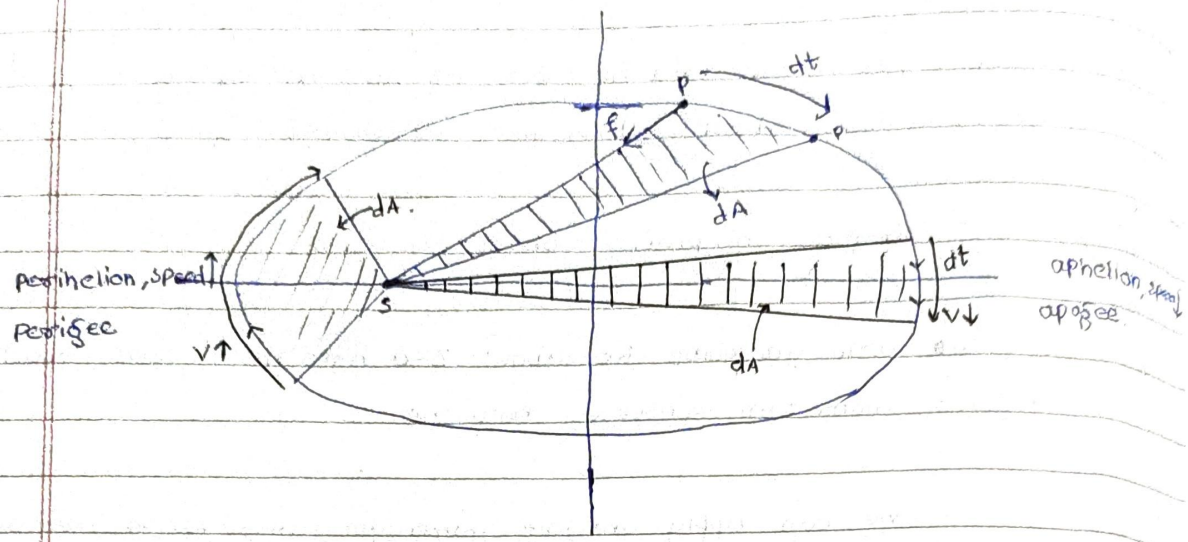
$$\frac{dA}{dt} = \frac{rv\sin\theta}{2} = \frac{L}{2m} = \text{constant.}$$

ex-



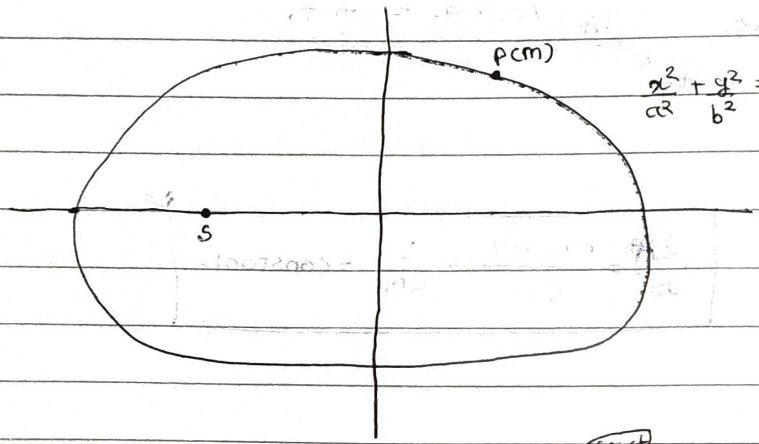
#4.

Kepler's 2nd law :-



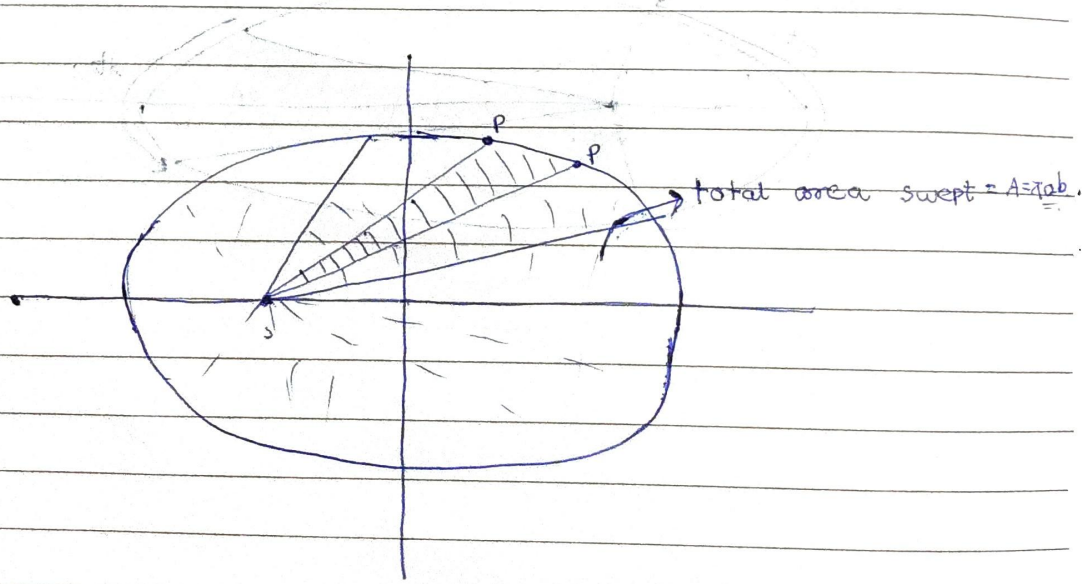
$$\text{areal velocity} = \frac{dA}{dt} = \frac{L}{2m}$$

Q-14



Angular momentum of P about sun is $\underset{\text{(const)}}{\underline{L}}$. Find time taken by planet to complete 1 revolution about sun.

Soln-



$$\frac{dA}{dt} = \frac{L}{2m}.$$

$$\Rightarrow \int_0^{\pi ab} dA = \frac{L}{2m}\int_0^T dt.$$

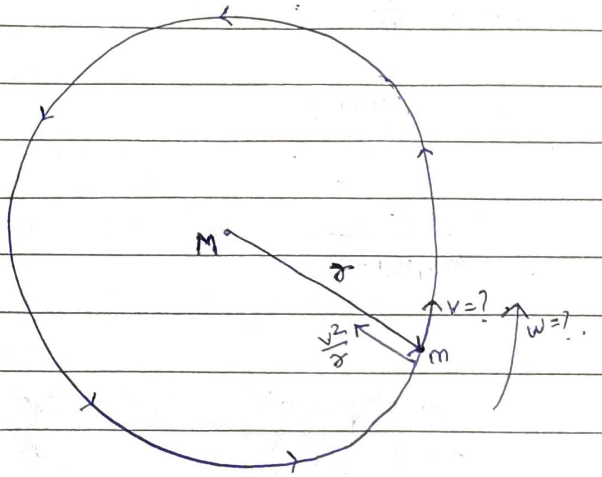
$$\Rightarrow \pi ab - 0 = \frac{L}{2m}[T-0].$$

$$\boxed{T = \frac{2\pi mab}{L}}$$

Rem! ae a formula. If u can.

③ Kepler's third law :-

(i)



N.L.M, $\frac{GMm}{r^2} = \frac{mv^2}{r}$

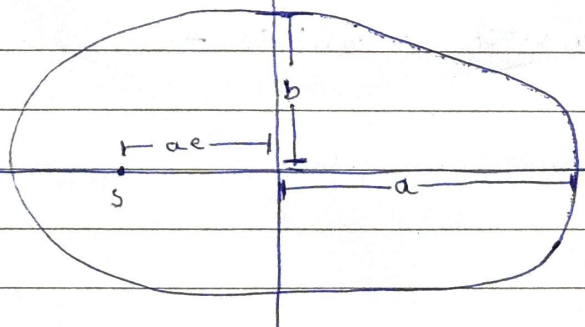
$$v = \sqrt{\frac{GM}{r}}$$

$$\omega = \frac{v}{r} = \sqrt{\frac{GM}{r^3}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r^3}{GM}}$$

$$\boxed{T^2 \propto r^3} \quad 3^{rd} \text{ Law.}$$

(ii)



$T^2 \propto a^3$

(Time period)^2 $\propto$ (Semi major axis)^3

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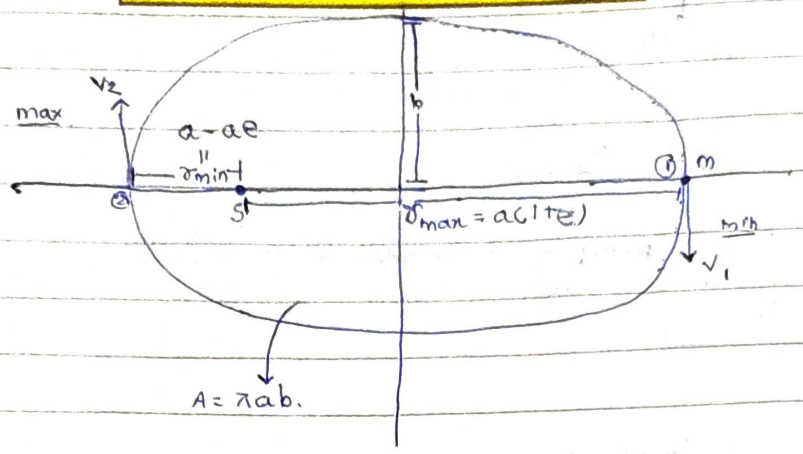
proof :-



$$\pi ab = \frac{L}{2m} . T$$

$$T = \frac{2\pi mab.}{L}$$

$$= \frac{2\pi ma}{mv_1 a(1+e)}$$

$$= \frac{2\pi m a^2 \sqrt{1-e^2}}{mv_1 a (1+e)}$$

$$T = \frac{2\pi a}{v_1} \sqrt{\frac{1-e}{1+e}}$$

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

$$\frac{b^2}{a^2} = 1 - e^2$$

$$b = a\sqrt{1-e^2}$$

Jaa sahide

bcz ek force lag raha he bcoz sun se pass horaha he ($\tau = 0$)

if $\tau = 0$ then angular mom^ conserve

① / ② Angular momentum conservation theorem.

$$mv_1 a(1+e) = mv_2 a(1-e)$$

$$v_1(1+e) = v_2(1-e) \quad - ①$$

$L = mvr \sin\theta$

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near no ext. force & gravitational force is a conser force.

Total mechanical energy conservation [TMEC] :-

$$\frac{1}{2}mv_1^2 - \frac{GMm}{a(1+e)} = \frac{1}{2}mv_2^2 - \frac{GMm}{a(1-e)}$$

{ bet sun and planet
{ M - mass of planet.

$$v_1^2 - \frac{2GM}{a(1+e)} = \frac{v_1^2(1+e)^2}{(1-e)^2} - \frac{2GM}{a(1-e)}$$

$$\frac{v_1^2(1+e)^2}{(1-e)^2} - v_1^2 = \frac{2GM}{a}\left\langle \frac{1}{1-e} - \frac{1}{1+e} \right\rangle$$

$$v_1^2\left\langle \frac{(1+e)^2 - (1-e)^2}{(1-e)^2} \right\rangle = \frac{2GM}{a}\left\langle \frac{2e}{(1+e)(1-e)} \right\rangle$$

$$\frac{v_1^2 4e}{(1-e)} = \frac{4GMe}{a(1+e)}$$

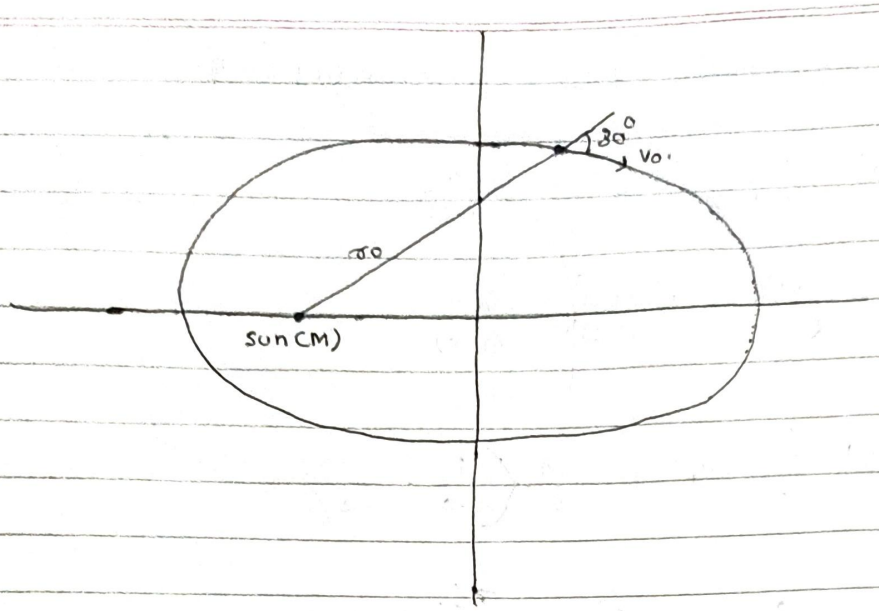
$$v_1 = \sqrt{\frac{GM}{a}\frac{(1-e)}{(1+e)}}$$

$$T = \frac{2\pi a}{\sqrt{\frac{GM}{a}\frac{(1-e)}{(1+e)}}}\sqrt{\frac{1-e}{1+e}} = \frac{2\pi}{\sqrt{GM}}a^{3/2}$$

$$\boxed{T = \frac{2\pi}{\sqrt{GM}}a^{3/2}} \qquad T \propto a^{3/2}$$

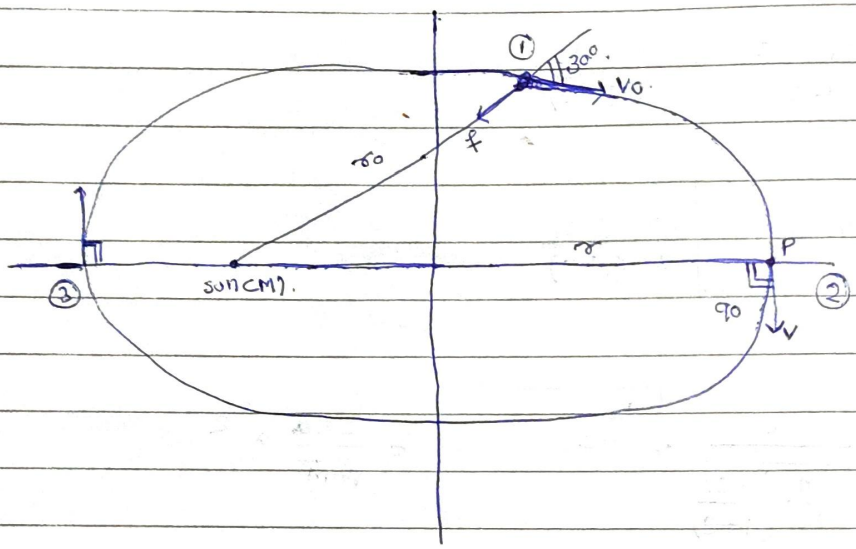
$$\boxed{T^2 = \frac{4\pi^2}{GM}a^3}$$ Real

Q-15



Find max^m and min^m separation b/w sun and planet.

Sol^n-



A.M.C.T :-

$$m v_0 r_0 \sin 30^\circ = m v r \sin 90^\circ$$

$$v r = \frac{v_0 r_0}{2} \quad \text{— (1)}$$

TMEC (system have no external force, but gravitational force is conservative force).

$$\frac{1}{2} m v_0^2 - \frac{GMm}{r_0} = \frac{1}{2} m v^2 - \frac{GMm}{r} \quad \text{— (2)}$$

From eq-① & ②.

we dont have intrest in finding 'v', so eliminate it to find r. if we have been asked to find v we eleminate r and find it.

$$v_0^2 - \frac{2GM}{r_0} = \frac{v_0^2 r_0^2}{4r^2} - \frac{2GM}{r} \qquad [\times r^2]$$

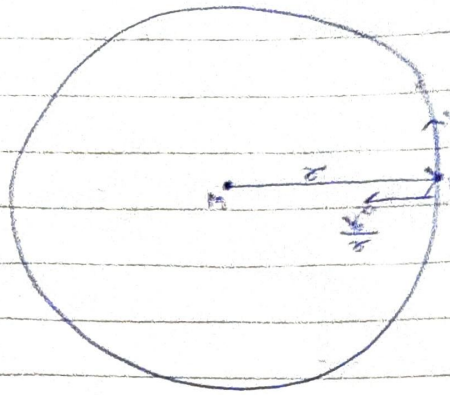
$$r^2 v_0^2 - \frac{2GM}{r_0} r^2 = \frac{v_0^2 r_0^2}{4} - 2GMr$$

$$r^2 \left\langle \frac{2GM}{r_0} - v_0^2 \right\rangle - 2GMr + \frac{v_0^2 r_0^2}{4} = 0.$$

$$r = \frac{2GM \pm \sqrt{4G^2M^2 - 4\left(\frac{2GM}{r_0} - v_0^2\right)\frac{v_0^2 r_0^2}{4}}}{2\left\langle \frac{2GM}{r_0} - v_0^2 \right\rangle}$$

Hear we got two value of 'r' the small-value is r_{min}. & bigger value is r_{max}. bcz. if we write eq" bet" ① & ③ then they are more or less same as ① & ②.

Motion of satelite :-



r = distance from core of planet.

$$\frac{GMm}{r^2} = \frac{mv_0^2}{r}$$

$$\text{orbital velocity} = v_0 = \sqrt{\frac{GM}{r}} = \frac{v_{escape}}{\sqrt{2}} = 0.707 v_{es}$$

$$\omega = \frac{v_0}{r} = \sqrt{\frac{GM}{r^3}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r^3}{GM}}$$

Some interesting things :-

* Manmade satelite have many uses,

Like- ① some satelite having cameras, which is used as spying, mausam ki jan kari.

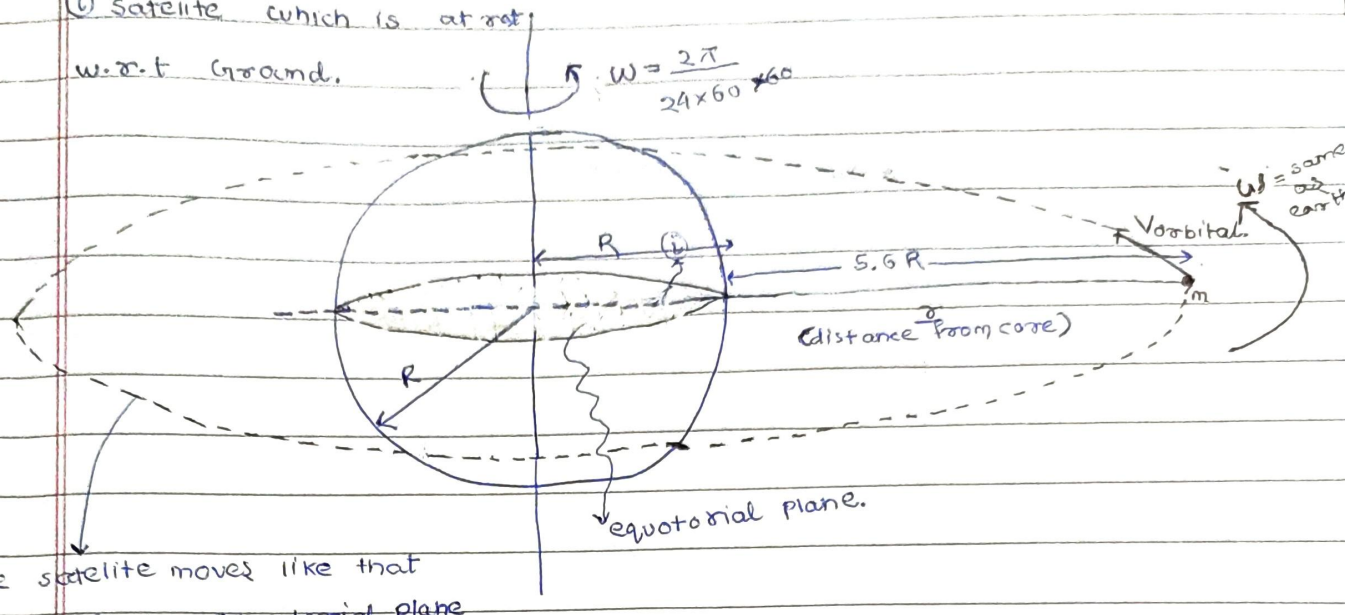
② communication satelite., for signal transfer.

Etc. etc.

* Earth rotate from west to east.

Geostationary satelite :-

① Satelite which is at rest w.r.t Ground.



① ---- blue dotted line is equatorial line.

the satelite moves like that it moves in equatorial plane of earth.

$$\omega = \sqrt{\frac{GM}{r^3}} = \frac{2\pi}{24 \times 60 \times 60}$$

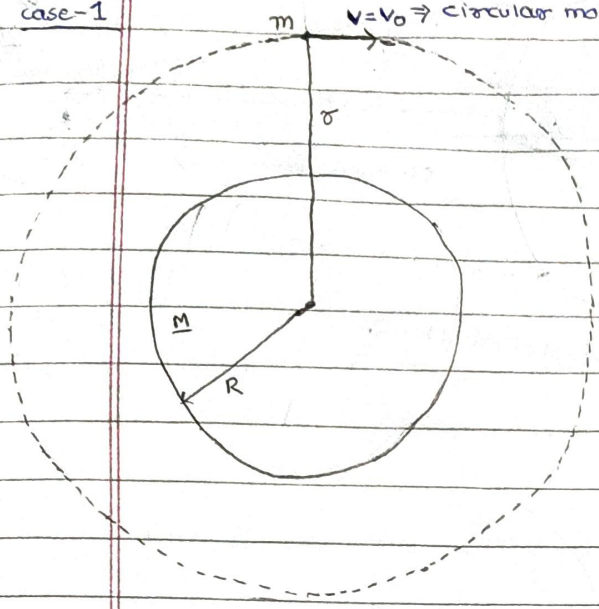
by putting value & equating.

$r = 6.6(R)$ → Radius of earth = 6400 km.

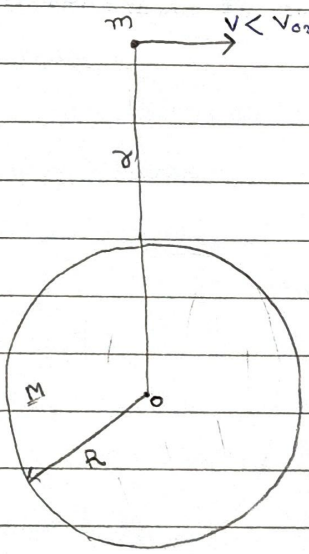
$$V_{orbital} = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{6.6R}}$$

important notes :-

case-1



case-2

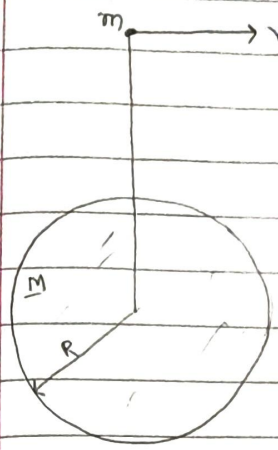


Note - for circular motion.
v should excactly equal to $v_{orbital}$.

- if $v \neq v_{orbital}$ motion is not circular.

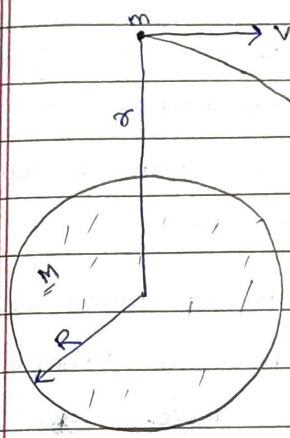
→ 'O' is one focus of elipse.

Case - III



$V_0 < V < V_e \Rightarrow$ elliptical motion (Bounded space)

Case - IV.

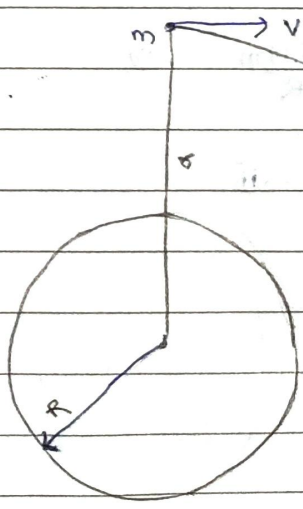


$TME = \frac{1}{2}m\,2GM - \frac{GMm}{r} = 0$

∞

∞ tak.
circle ✗.
ellipse ✗.
Parabola ✓.
hyperbola ✓.

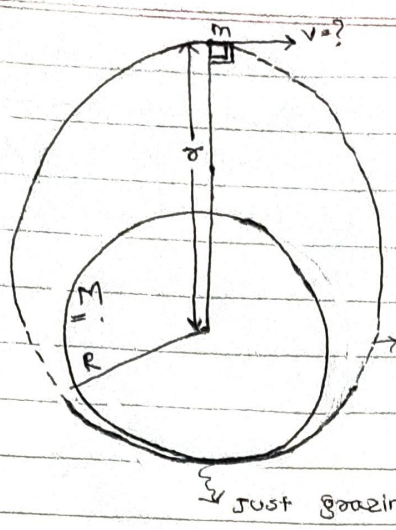
Case - V.



TME = +ve.

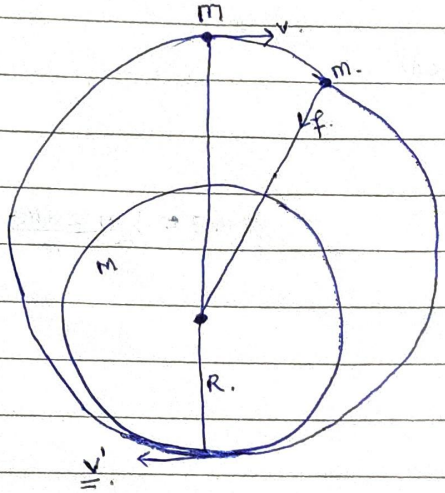
∞

Q-16



find v.

Soln.



Satelite par ek bata force lagraha he 'gravitational force' jo hamesha earth ke center se pass karega.

To earth ke center ke about planet par jo torque ayega vo zero ayega. ✱.

if $\tau = 0 \Rightarrow$ Angular momentum conserved.

AMCT.

$$L_i = L_f.$$

$\vec{L} = m(\vec{v} \times \vec{r})$.

$$\therefore mvr = mv'R. \quad -(i) \qquad \Rightarrow vr = v'R. \quad -(1)$$

TMEC.

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = \frac{1}{2}mv'^2 - \frac{GMm}{R}$$

$$\frac{v^2}{2} - \frac{GM}{r} = \frac{v'^2}{2} - \frac{GM}{R} \quad -(ii)$$

$$v^2 = \frac{2GM}{r} = \frac{v^2r^2}{R^2} - \frac{2GM}{R}$$

$$v^2\left(\frac{r^2}{R^2}-1\right) = 2GM\left(\frac{1}{R}-\frac{1}{r}\right)$$

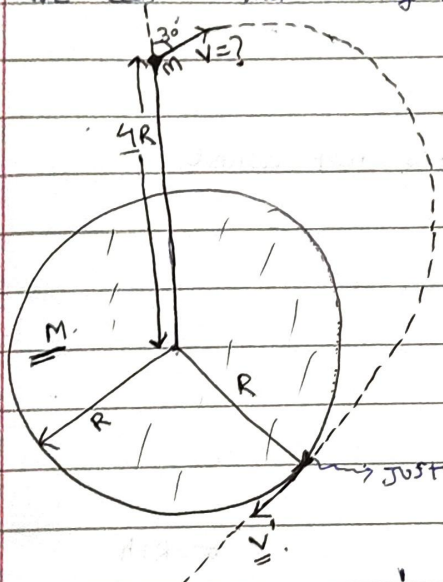
$$\Rightarrow v^2 = 2GM\left(\frac{r-R}{Rr}\right)\left(\frac{R^2}{r^2-R^2}\right).$$

$$\Rightarrow v = \sqrt{\frac{2GM(r-R)R^2}{R\,r\,(r^2-R^2)}}$$

$$\text{Sc. } v = \sqrt{\frac{2GMR}{r(r+R)}} \quad \text{And.}$$

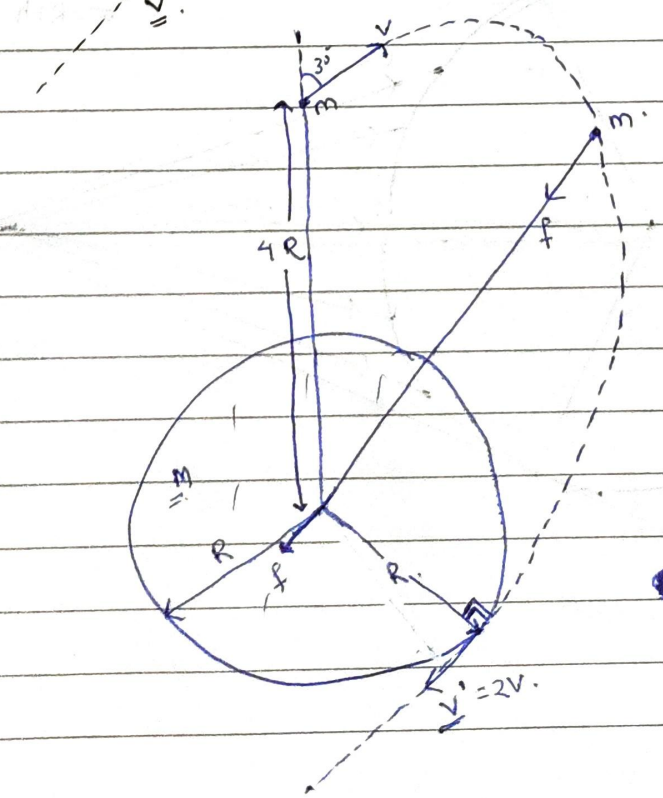
we can find v' by putting value of v in eq (1). thats all.

(Q-17)



find $\underline{v}$.

Soln-



same as (*).

if $\tau = 0 \Rightarrow$ Angular momentum conserved.

AMOT.

$L_i = L_f$

$mv4R\sin 30° = mv'R\sin 90°$.

angle How?

$2v = v' \quad -(1)$

TMEC

(Mechanical energy ka loss ka hazard to hehin aa raha he, Na koi external agent ka workdone, Na koi resistive forces ka workdone, to hum total mechanical energy conserve kar sakte hain)

$K_i + U_i = K_f + U_f$

$\frac{1}{2}mv^2 - \frac{GMm}{4R} = \frac{1}{2}mv'^2 - \frac{GMm}{R}$

$\Rightarrow v^2 - \frac{GM}{2R} = v'^2 - 2\frac{GM}{R}$

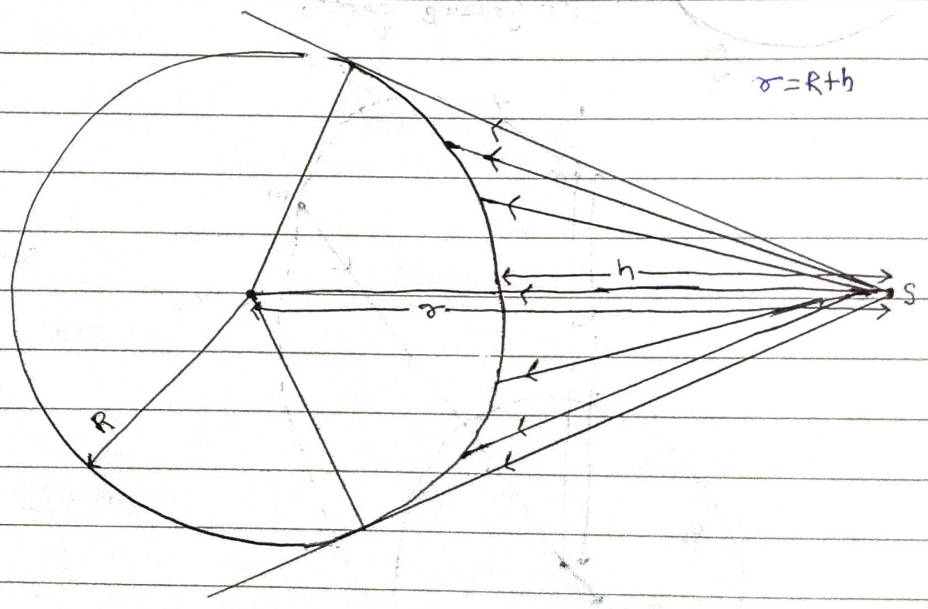
$\Rightarrow v^2 - \frac{GM}{2R} = 4v^2 - 2\frac{GM}{R}$

$\Rightarrow 3v^2 = \frac{GM}{R}\frac{3}{2}$

$\Rightarrow v = \sqrt{\frac{GM}{2R}}$ Ans.

$v' = 2v = 2\sqrt{\frac{GM}{2R}}$, although it is not asked.

Region covered by a satellite :-



classmate
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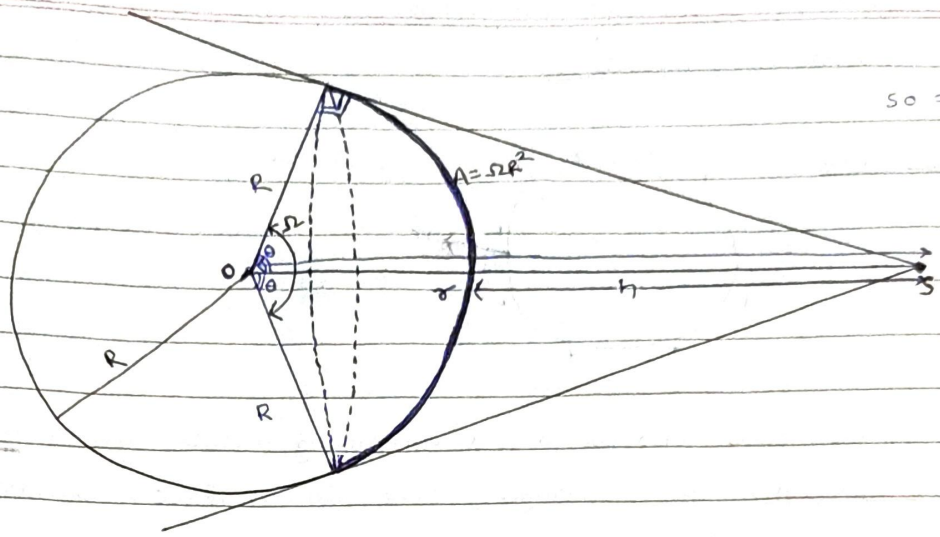


So $= r = R + h$

$$\cos\theta = \frac{R}{r} \Rightarrow \theta = \cos^{-1}\frac{R}{r}$$

Solid Angle $= \Omega = 2\pi\left\langle 1 - \cos\theta \right\rangle$

$$= 2\pi\left\langle 1 - \frac{R}{r} \right\rangle.$$

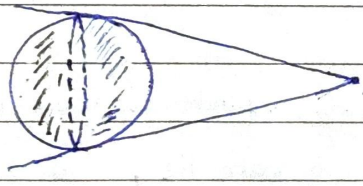
Area receiving signal $= A = \Omega R^2 = \frac{2\pi(r-R).R^2}{r} = \frac{2\pi h.R^2.}{(R+h)}$

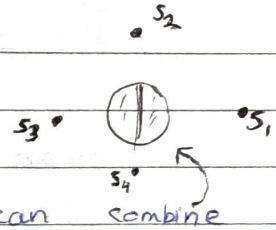
$$A = \frac{2\pi R^2 \, h}{R+h.} \quad \text{or} \quad A = \frac{2\pi (r-R) R^2.}{r.}$$

$$A = \frac{2\pi R^2.}{\left(\frac{R}{h} + 1\right)}, \quad h\uparrow \Rightarrow A\uparrow,$$

NOTE- mathematically maximum area coved by Satelite $A = 2\pi R^2$.

Visually-



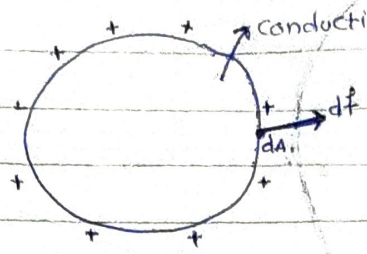


but to provide signal to other end we can combine satelite tneg send signal each other.

JEE main ✗ JEE Advance / olympiad ✓

Gravitational Pressure :-

electrostat concept rewind :-

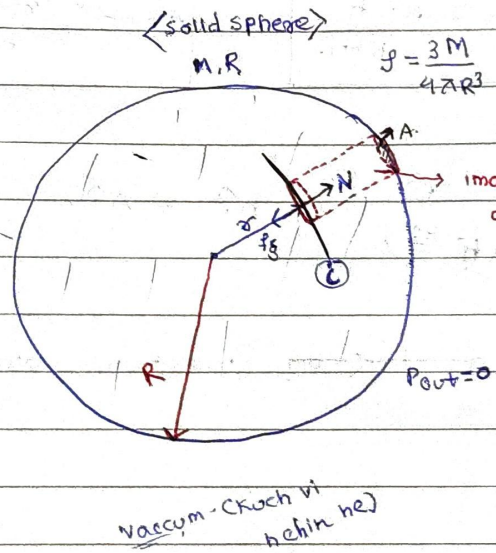


if we gave charge to a conducting metal. due to repulsion phenomena, all charge comes to surface. surface pe aane ke bad bhi bo ek dusre ko repeal kar rahe hain, to bahar ki taraf ek force lag raha he.

that force per unit area is electric pressure.

$$P = \frac{df}{dA} = \frac{\sigma^2}{2\varepsilon_0}$$

Now in gravitation :-



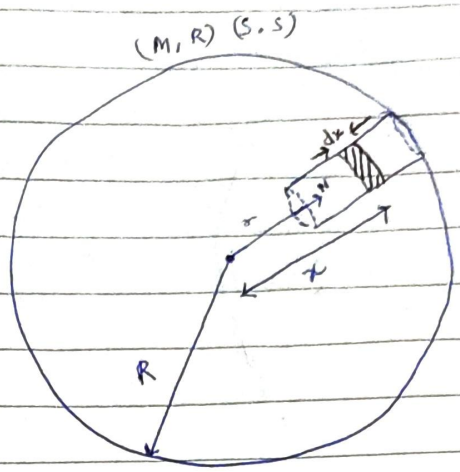
vaccum - (kuch vi nehin he)

Rem!
* in case of charge charges are repel each other to go away but in case of (gravitation) masses they attract isiliey tendency bahar jane ki bajaye andar ane ki hogi, i mean ki same ke same mass. andar ane chahenge.

(i) The black line - imaginary cylindrical mass per andar ki taraf gravitational Force lag raha he, to imag^n cylind^n mass andar ana chahega, but the black line (rock, crust) koi na koi Normal force lagaayega.

$$\Rightarrow \frac{N}{A} = P$$

Now we find 'N'.



force on dx due earth = $df = (\rho A dx)\left(\frac{GMx}{R^3}\right)$

" " Imagn cylinder " $= f = \int df = \frac{\rho AGM}{R^3}\int_r^R x\,dx$

$$f = \frac{\rho AGM}{R^3}\left(\frac{x^2}{2}\right)_r^R$$

$$f = N = \frac{\rho AGM}{R^3}\left\langle\frac{R^2 - r^2}{2}\right\rangle$$

Jitna force andar ki taraf he hiche bale chatan ne utna normal force Laga dega.

$$P = \frac{N}{A} = \text{Gravitational pressure} = \frac{\rho GM}{2R^3}\langle R^2 - r^2\rangle$$

or

$$P = \frac{GM^2 3}{2R^3 4\pi R^3}\langle R^2 - r^2\rangle$$

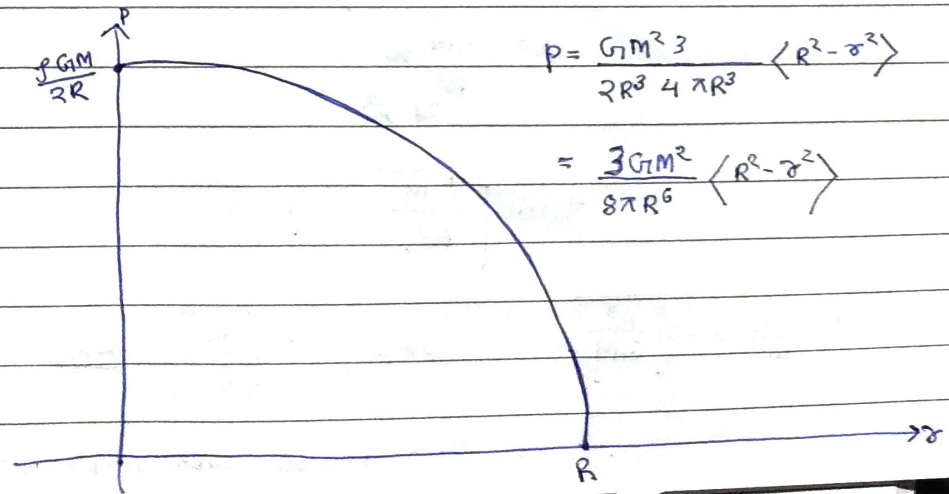
$$= \frac{3GM^2}{8\pi R^6}\langle R^2 - r^2\rangle$$

P ka power = 1

r ka " = 2

r ka coff = -ve

Parabola math open downward.



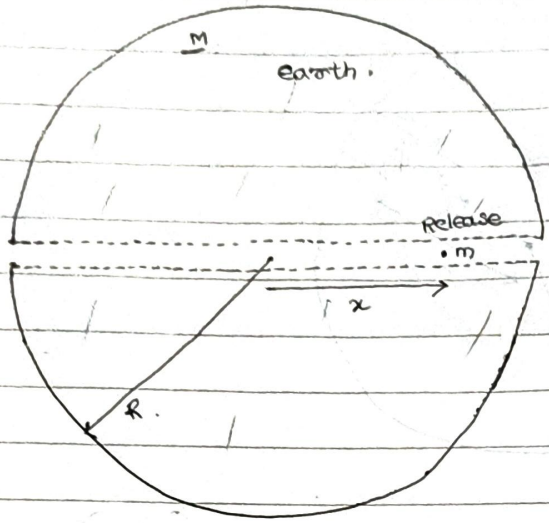
H.W sheet.

#5

Examples :-

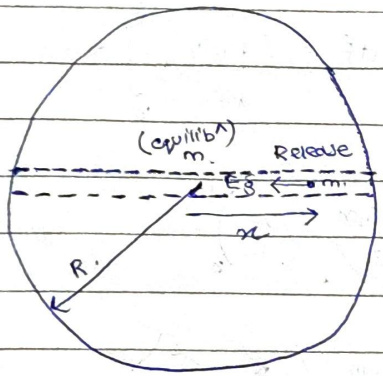
<Uniformly distributed>

Q-18



(i) Prove that m will perform SHM also find time period of SHM.

Soln-



at core-

$f = m \times 0 = 0$.

at x' dist -

$f = mE_g$

$= \frac{m G M x}{R^3}$

$= \left(\frac{mGM}{R^3}\right) x$

force opposite to displacement. & $f \propto x$ $\Rightarrow$ SHM proved.

$$f = ma = \left(\frac{mGM}{R^3}\right) x$$

$$a = \frac{GM}{R^3} x$$

$\downarrow \omega^2$

$$\Rightarrow \omega = \sqrt{\frac{GM}{R^3}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{R^3}{GM}} \quad \text{Or} \quad T = 2\pi \sqrt{\frac{R}{g}} \quad \text{Ans.} \quad \left\langle g = \frac{GM}{R^2} \right\rangle$$

S.H.M ka time period kabhi Amplitude par depend nahin karta hai.

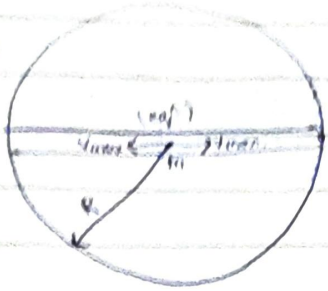
(A)

(B) Find v_{max} of mass m if it is released from surface.

(C) Find v of mass m when it is at a distance $R/2$ from centre.

Sol^n

(A)



TMEC

$K_i + U_i = K_f + U_f$

$$0 + m\left\langle \frac{-GM}{R} \right\rangle = \frac{1}{2}mv_{max}^2 + m\left\langle \frac{-GM}{2R^3}(3R^2 - R^2) \right\rangle$$

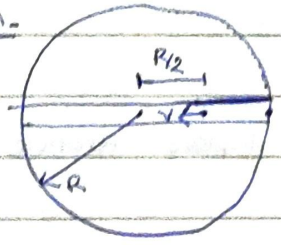
$$m\left\langle -\frac{GM}{R} \right\rangle = \frac{1}{2}mv_{max}^2 + m\left\langle -\frac{3}{2}\frac{GM}{R} \right\rangle$$

$$\frac{1}{2}\frac{GM}{R} = \frac{1}{2}v_{max}^2$$

$$\Rightarrow v_{max} = \sqrt{\frac{GM}{R}} \quad \text{Ans.}$$

by S.H.M concept - $v_{max} = A\omega = R\sqrt{\frac{GM}{R^3}} = \sqrt{\frac{GM}{R}}$

Sol^n (B) Sol^n -



TMEC.

$K_i + U_i = K_f + U_f$

$$0 + m\left\langle \frac{-GM}{R} \right\rangle = \frac{1}{2}mv^2 + m\left\langle -\frac{GM}{2R^3}\left(3R^2 - \frac{R^2}{4}\right) \right\rangle$$

$$m\left\langle -\frac{GM}{R} \right\rangle = \frac{1}{2}mv^2 + m\left\langle -\frac{GM}{2R^3}\left(\frac{11R^2}{4}\right) \right\rangle$$

$$\frac{v^2}{2} = \frac{GM}{R}\left\langle \frac{11}{8} - 1 \right\rangle$$

$$v^2 = \frac{GM}{R} \times \frac{3}{4} \quad \Rightarrow v = \sqrt{\frac{3}{4}\frac{GM}{R}} \quad \text{Ans.}$$

Ye bala Particle ka S.H.M kaise nahi hoga, by using concepts of S.H.M-,

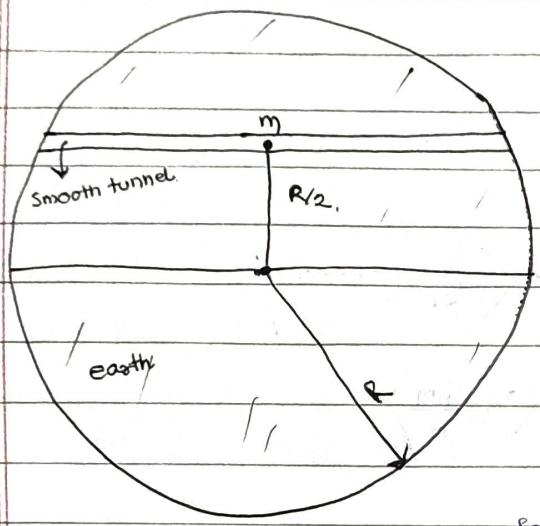
Velocity of particle, if it is x distance from equilibrium $V = \omega\sqrt{A^2 - x^2}$

So,

$$V = \omega\sqrt{A^2 - x^2}$$

$$= \sqrt{\frac{GM}{R^3}}\sqrt{R^2 - R^2/4} = \sqrt{\frac{GM}{R^3}} \times \sqrt{\frac{3R^2}{4}} = \sqrt{\frac{3}{4}\frac{GM}{R}} \quad \text{Ans.}$$

Q-19

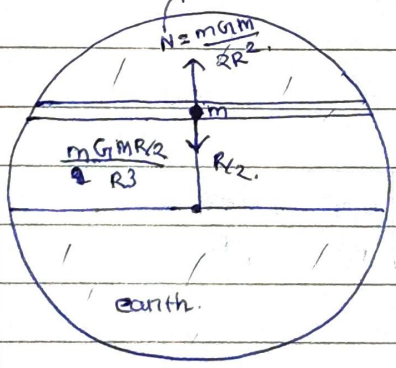


prove that m is displaced inside the tunnel it will perform S.H.M.

Also find time period of S.H.M.

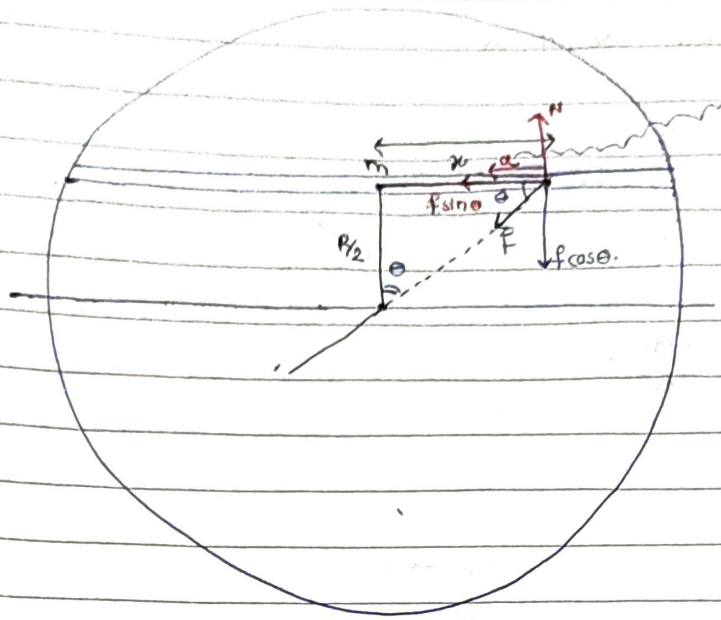
Soln.

Normal force by surface of tunnel, to balance gravitat" force on 'm'.



Initially particle m is at equilibrium.

Now to prove S.H.M we have to displace particle slightly.



→ SHM ka 1st condition fulfilled. bcz displacement & f_net opposite. – (i).

$$f = \frac{m\,GM\sqrt{\frac{R^2}{4}+x^2}}{R^3}.$$

$$f_{restoring} = f\sin\theta = \frac{GMm\sqrt{\frac{R^2}{4}+x^2}}{R^3} \times \frac{x}{\sqrt{\frac{R^2}{4}+x^2}}$$

$$= \frac{GMm}{R^3}\,x.$$

(i) & $f_{restor} \propto x$ ⇒ S.H.M proved.

$$ma = \frac{GMm}{R^3}\,x.$$

$$\Rightarrow a = \underbrace{\frac{GM}{R^3}}_{\omega^2}\,x$$

$$\omega = \sqrt{\frac{GM}{R^3}} \Rightarrow T = 2\pi\sqrt{\frac{R^3}{GM}}$$ Ans.

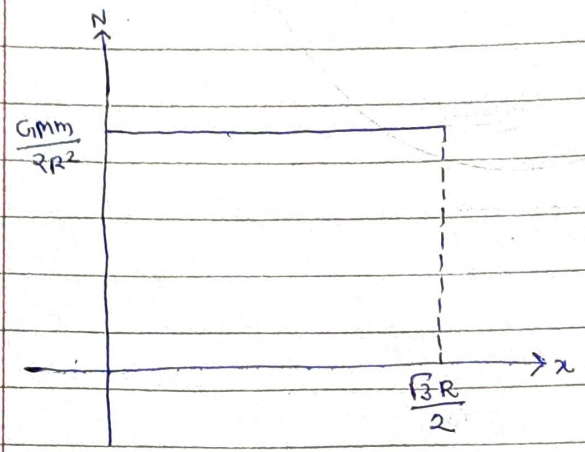
$$= 2\pi\sqrt{\frac{R}{g}}$$

(iii) draw graph b/w N and x.

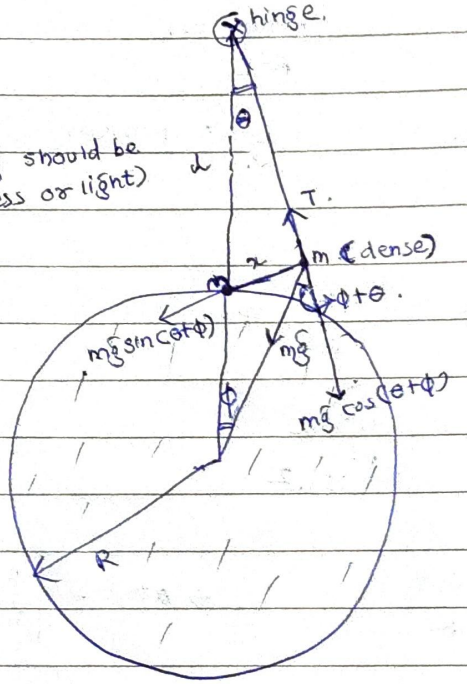
Soln - $N = f\cos\theta$

$$= \frac{mGM\sqrt{\frac{R^2}{4}+x^2}}{R^3} \times \frac{R}{2\sqrt{\frac{R^2}{4}+x^2}}$$

$$= \frac{GMm}{2R^2}$$



Very long simple pendulum :-



Approximation:- angular displace^ should be very small. if angular displa^ is big we can't solve it.

→ the displace(x) is very small if it is big gravitational pull become change.

earth pull kaanti he aphi core ki taraph.

$\sin x = x$ (small angle Approx).

$f_{restoring} = mg\sin(\theta+\phi)$

$= mg(\theta+\phi)$ (small angle approxi^).

classmate

Gate Page

$F_{restor} = mg\left(\frac{x}{L} + \frac{x}{R}\right)$ (if it is small^ly that we can take it as arc).

$\theta = \frac{\text{Arc length}}{\text{Radius}}$.

$F_{rest.} = mg\left\langle \frac{1}{L} + \frac{1}{R} \right\rangle x \Rightarrow$ SHM.

∴ Force, Angular SHM kare nahe the but hum uska Linear type approximation manke force find kiye.

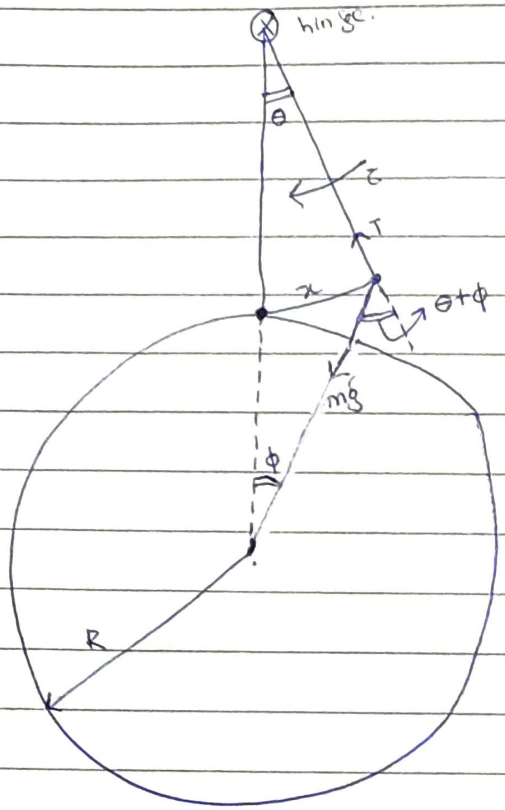
$ma = mg\left\langle \frac{1}{L} + \frac{1}{R} \right\rangle x$

$\Rightarrow a = g\left(\frac{1}{L} + \frac{1}{R}\right)x$

ω^2

$\Rightarrow \omega = \sqrt{g\left\langle \frac{1}{L} + \frac{1}{R} \right\rangle}$ if $L \to \omega$ $\omega = \sqrt{g/R}$.

& $T = \frac{2\pi}{\omega}$ Put value ....



hinge ke about particle pe torque.

Tension ka = 0, perpendicular force pass through hinge.

$mg\cos(\theta+\phi)$ ka bhi = 0

$mg\sin(\theta+\phi)$ ka torque ayega.

$\tau = mg\sin(\theta+\phi)\,L$.

$\tau = mg(\theta+\phi)L$

① Angular displacement & τ are opposite. Now we prove $\tau \propto \theta$.

τ

$x = l\theta = R\phi$

$\phi = \frac{l\theta}{R}$

$\tau = mg\left(\theta + \frac{l\theta}{R}\right)l$.

$= mg\left\langle l + \frac{l^2}{R}\right\rangle\theta$. $\Rightarrow$ Angular SHM proved.

$I\alpha = (mR)\alpha = mg\left\langle l + \frac{l^2}{R}\right\rangle\theta$.

$\Rightarrow \alpha = g\left\langle \frac{1}{l} + \frac{l}{R}\right\rangle\theta$.

$= \omega^2$.

$\omega = \sqrt{g\left\langle \frac{1}{l} + \frac{1}{R}\right\rangle}$

$K = \frac{1}{2}mv^2$

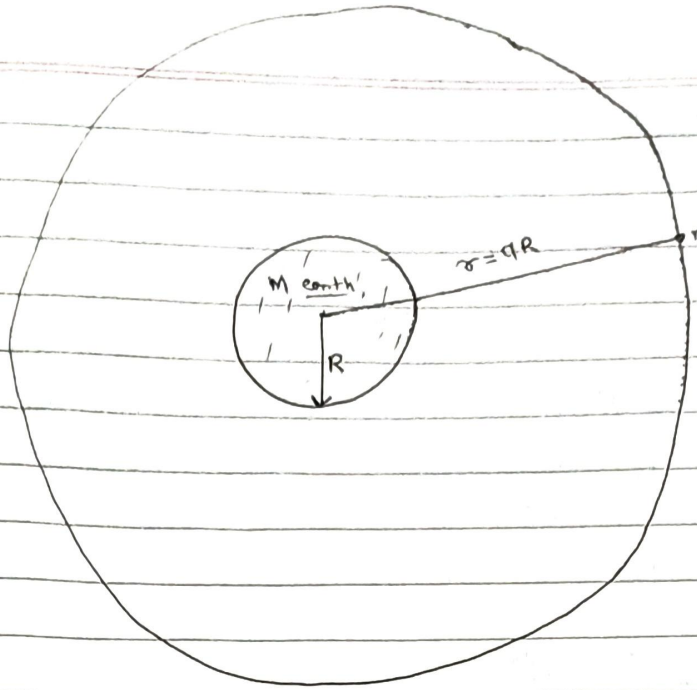
$K = \frac{1}{2}m\frac{GM}{r}$

$dK = \frac{GMm}{2}\left(\frac{-1}{r^2}dr\right)$

dr = -ve so dK = +ve

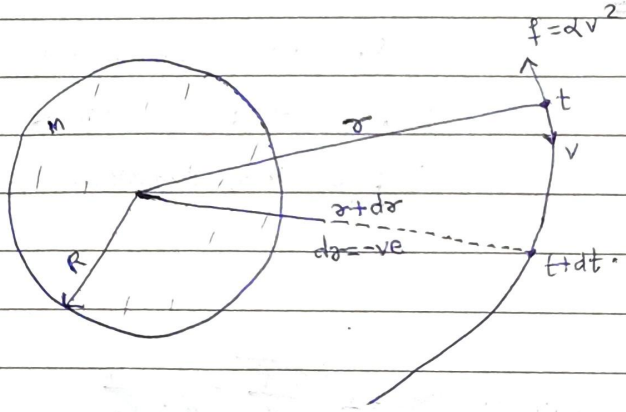
* Y

(P-20)



Satelite of mass m is moving in orbit of radius qR. it experience a resistive force $f = \alpha v^2$, due to which it keep on moving on a spiral path and strike the plannet surface find time taken by the satelite to strike the plannet.

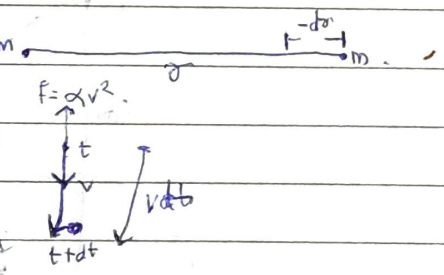
Sol^-



TME C - ✗ bcz resist^ is dissipate energy.

WET.

$dw_g + dw_R = dK.$



$\Rightarrow \frac{G_1 Mm}{r^2}(-dr)\cos 0^\circ + (\alpha v^3)v\,dt\cos\pi = dK$

$W = f s \cos\theta$

$\cos\pi = -1$

$\Rightarrow -\frac{G_1 Mm}{r^2}(dr) - \alpha v^3 dt = -\frac{G_1 Mm}{2r^2}dr$

$K = \frac{1}{2}mv^2.$

$$\Rightarrow \quad -\alpha v^3 dt = \frac{GMm\,dr}{2r^2}$$

$$\Rightarrow \quad -\alpha \left(\frac{GM}{r}\right)^{3/2} dt = \frac{GMm\,dr}{2r^2}$$

$$\Rightarrow \quad -\alpha \sqrt{GM}\, dt = \frac{m}{2\sqrt{r}}\, dr$$

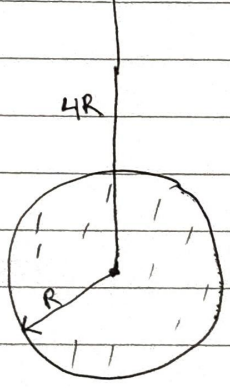
$$\Rightarrow \quad \frac{-\alpha 2\sqrt{GM}}{m} \int_0^t dt = \int_{9R}^{R} r^{-1/2}\, dr.$$

$$\Rightarrow \quad \frac{-2\alpha\sqrt{GM}}{m}\, t = \left(2r^{1/2}\right)_{9R}^{R}$$

$$\Rightarrow \quad \frac{-2\alpha\sqrt{GM}}{m}\, t = 2\left\langle \sqrt{R} - 3\sqrt{R} \right\rangle$$

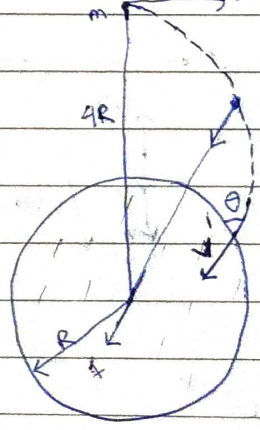
$$\Rightarrow \quad t = \frac{2\sqrt{R}\, m}{\alpha\sqrt{GM}} \qquad \text{Ans.}$$

(Q-2) (satelite) m $\rightarrow V = \frac{V_{orbital}}{2}$



find the angle at which this satelite will strike the earth's surface.

Soln-



$$V = \frac{V_{orbital}}{2} = \frac{1}{2}\sqrt{\frac{GM}{4R}} = \frac{1}{4}\sqrt{\frac{GM}{R}}$$

$V < V_{orb}$

$\Downarrow$

elliptical motion.

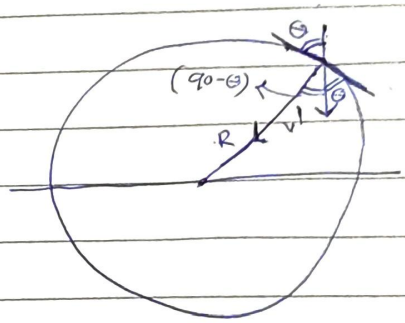
IMEC: <No Resistive force, No Non conservative force are there>.

$$\frac{1}{2}mv^2 + \frac{1}{16}\frac{GMm}{R} + m\left\langle\frac{-GM}{4R}\right\rangle = \frac{1}{2}mv'^2 + m\left\langle\frac{-GM}{R}\right\rangle \quad -①$$

$L_i = L_f$ <Pure ke pure satelite per ek hi force h.e. vo hamesha earth ke center se gujar rahe he. then earth ke about satelite per kisne bala torque zero he. ⇒ Angular momentum conserved>.

$$m\frac{1}{4}\sqrt{\frac{GM}{R}} \times 4R = mv'R\sin(90-\theta)$$

$$\sqrt{\frac{GM}{R}} = v'\cos\theta \quad -②$$



from eq-①

$$2\left(\frac{GM}{32R} - \frac{GM}{4R} + \frac{GM}{R}\right) = v'^2 \quad \Rightarrow v'^2 = 2\left\langle\frac{1-8+32}{32R}\right\rangle GM$$

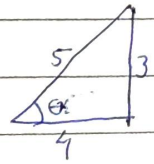
$$\Rightarrow v'^2 = \left(\frac{25}{16R}\right)GM.$$

$$\Rightarrow v' = \frac{5}{4}\sqrt{\frac{GM}{R}} \quad -①.$$

Put in eq ②.

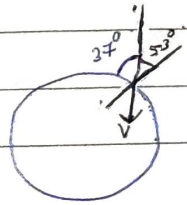
$$\sqrt{\frac{GM}{R}} = \frac{5}{4}\sqrt{\frac{GM}{R}}\cos\theta \quad \Rightarrow \cos\theta = \frac{4}{5}.$$

$\Rightarrow \theta = 37°$. Ans.

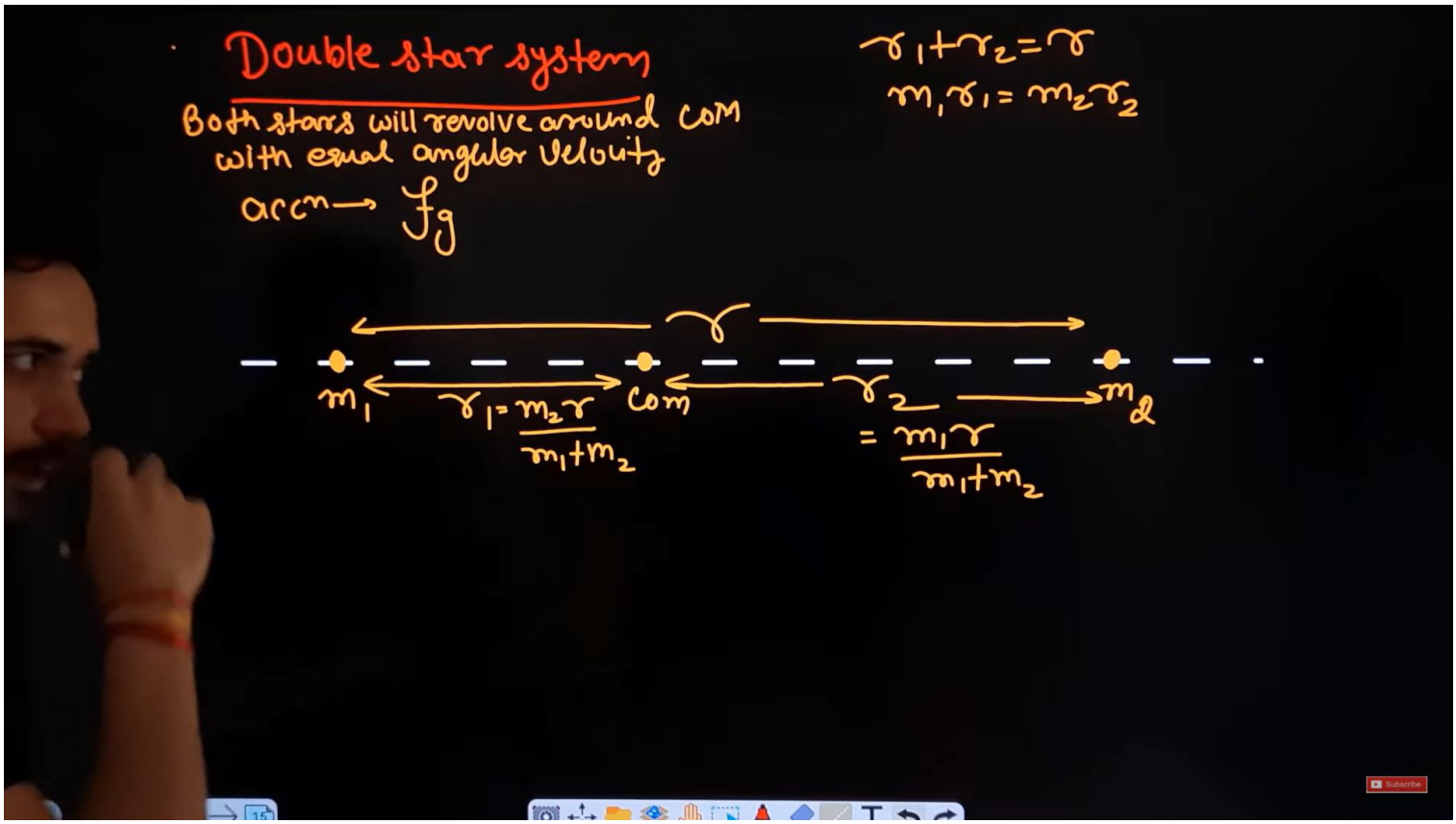


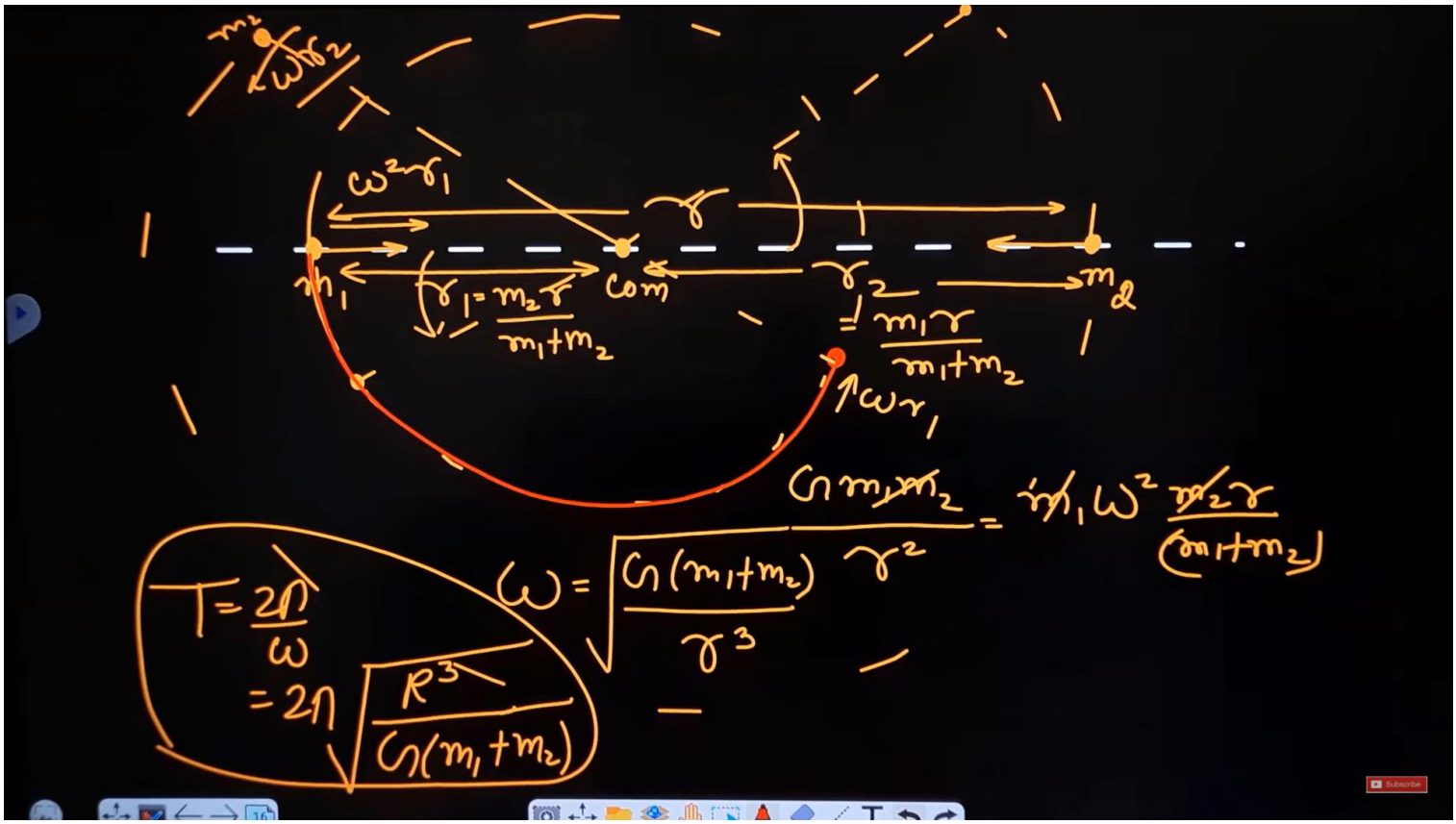
surface ke sath 37°.

vertical " " 53°.



Happy ending.


Double star system
Both stars will revolve around COM with equal angular velocity
arcn → f_g
$r_1 + r_2 = r$
$m_1 r_1 = m_2 r_2$
m_1
$r_1 = \frac{m_2 r}{m_1 + m_2}$
COM
$r_2 = \frac{m_1 r}{m_1 + m_2}$
m_2


m_2
$\omega^2 r_2$
T
$\omega^2 r_1$
r
m_1
$r_1 = \frac{m_2 r}{m_1+m_2}$
com
$r_2 = \frac{m_1 r}{m_1+m_2}$
m_2
ωr_1
$\frac{G m_1 m_2}{r^2} = m_1 \omega^2 \frac{m_2 r}{(m_1+m_2)}$
$\omega = \sqrt{\frac{G(m_1+m_2)}{r^3}}$
$T = \frac{2\pi}{\omega}$
$= 2\pi \sqrt{\frac{R^3}{G(m_1+m_2)}}$